

Labor Market Signals: The Role of Large Language Models*

Kian Abbas Nejad Giuseppe Musillo

Till Wicker Niccolò Zaccaria

Abstract

Large Language Models (LLMs) are transforming the labor market, including hiring decisions. This paper examines their impact on the signals job-seekers send to potential employers through two field experiments focusing on cover letters. We find that LLMs enhance signal quality, especially benefiting lower-quality applicants, yet these gains do not boost interview invitations because they are concentrated in standardized, less influential sections of the cover letter. When recruiters learn of LLM usage, they place greater value on high-quality, human-crafted letters. Hence, LLMs reduce the informativeness of signals, potentially increasing inefficiencies in labor market matching as estimated by a calibrated structural model.

Key words: Large Language Models; Cover Letters; Labor Market; Matching; Signaling.

JEL codes: C93; J24; O33; M51

***Abbas Nejad:** Tilburg University (e-mail: k.abbasnejad@tilburguniversity.edu); **Musillo:** Tilburg University (e-mail: g.musillo@tilburguniversity.edu); **Wicker:** Tilburg University (e-mail: t.n.wicker@tilburguniversity.edu) **Zaccaria:** Tilburg University (e-mail: n.zaccaria@tilburguniversity.edu). All authors contributed equally. We thank Francesco Capozza, Patricio Dalton, Mery Ferrando, Matteo Giugovaz, Djurre Holtrop, Boris van Leeuwen, Melika Liporace, Janneke Oostrom, David Schindler, Sjak Smulders, Has van Vlokhoven, Anna Salomons, Sigrid Suetens, and Burak Uras for their valuable comments. We also thank audience members at Tilburg University and the University of Tokyo for helpful discussions. Financial support from the Economics Department of Tilburg University is gratefully acknowledged. We thank OpenAI for supporting the textual analysis through its Researcher Access Program. The experiments received IRB approval (IRB FUL 2024-002 and TiSEM_RP1776, respectively) and were pre-registered (AEARCTR-0013355 and AEARCTR-0014784, respectively).

1 Introduction

The labor market is undergoing rapid transformations, driven by technological innovation and evolving workplace practices. Recent research highlights the impacts of these developments on modern labor dynamics (Deming and Kahn, 2018; Englmaier et al., 2024; Deming et al., 2025). Among the emerging technologies, the rise of Large Language Models (LLMs) promises to have substantial implications for the future of work, with at least 80% of the U.S. jobs expected to be affected, and 23% of U.S. employees already using LLMs at work (The White House, 2022; Eloundou et al., 2023; Bick et al., 2025). However, relatively little attention has been given to how LLMs can influence labor market matching: the process through which job-seekers are matched to firms.

Labor market matching heavily depends on signals, such as CVs and cover letters, since firms cannot directly assess a job-seeker’s productivity or fit. Cover letters, in particular, play a crucial role in many hiring processes, allowing job-seekers to showcase soft skills like motivation, communication abilities, and overall fit (ANP, 2023, 2024). 60-89% of recruiters across the US and the Netherlands require job-seekers to submit a cover letter, but the impact of LLMs on these signals is not clear ex-ante.¹ On one hand, LLMs may enhance job-seekers’ signals by improving writing quality (Noy and Zhang, 2023). On the other hand, they might render texts more formulaic and less personalized (Shanahan, 2024), reducing signal effectiveness. Despite uncertainty about whether LLMs help or harm employment prospects, more than half of job-seekers use them in applications (Criddle and Strauss, 2024). Do labor market signals written with LLM assistance help job-seekers get hired, and does this distort hiring decisions?

This paper answers this question through two pre-registered field experiments examining the impact of LLMs on labor market signals and analyzing both the perspectives of job-seekers and employers. The first experiment is conducted with job-seekers at two universities and recruiters from four multi-national companies with close to half a million employees. During the field experiment, some job-seekers were allowed to use ChatGPT when writing their cover letter, while others were not. Access to LLMs improves the quality of cover letters by more than 0.2 standard deviations, as evaluated by recruiters from

¹See Rendement (2024), Resume Genius (2024), and Zety (2024). The four most commonly cited pieces of information obtained from cover letters are: motivation for the job, language usage, match with the organization, and match with the job.

the four firms who were blind to the two treatments. In line with [Noy and Zhang \(2023\)](#), [Dell’Acqua et al. \(2023\)](#), [Caplin et al. \(2024\)](#), and [Brynjolfsson et al. \(2025\)](#), the positive treatment effects are stronger for lower-quality job-seekers, thus reducing the dispersion in quality of the cover letters, undermining their signaling value of the job-seeker’s true ability.

However, having access to LLMs does not increase the job-seeker’s likelihood of advancing to the next recruitment stage, an interview. This is because recruiters primarily care about two dimensions of the cover letter, neither of which improves as a result of access to LLMs. The first dimension is the job-seeker’s motivation. We find that ChatGPT improves less personalized and more generic sections, but falls short in enhancing more distinctive and individualized components such as the job-seeker’s motivation. Textual analysis of ChatGPT conversation histories reveals that job-seekers ask ChatGPT for help with their motivation multiple times; however, ChatGPT does not significantly improve this section.

The second dimension recruiters care about is the cover letter’s clarity, akin to recruiters in [Wiles et al. \(2025\)](#). Access to LLMs does not affect the perceived clarity of the cover letter (as evaluated by the recruiter) due to two competing forces. While LLMs enhance the cover letter’s clarity by reducing the frequency of grammatical mistakes by 39%, LLMs reduce clarity by *worsening* the text’s readability through using longer words. These two objective effects go in opposite directions, with LLMs not affecting the recruiter’s perception of the cover letters’ clarity. The lack of improvements in the motivation and clarity of the cover letter explains why, although LLMs improve the average quality of the cover letter, they do not increase the likelihood of the job-seeker being invited to an interview, the next stage in the recruitment process.

Recruiters’ perceptions of LLM usage can also affect the evaluation of labor market signals, with people having a preference for human over LLM-generated content ([Zhang and Gosline, 2023](#)). To understand this, we conducted a second experiment where we employed 401 recruiters to evaluate a subsample of cover letters from the first experiment. Some were explicitly informed about whether a cover letter had been written with LLM assistance, while others were only told that some letters had been written with the assistance of LLMs without knowing which ones. When recruiters were explicitly informed about LLM usage, they do not rate cover letters differently. However, there is substantial heterogeneity: while the evaluation of low and medium-quality cover letters do not differ systematically, recruiters evaluate high-quality cover letters written without LLM assistance more positively and are

more likely to invite the applicant to the next stage of the recruitment process.

Related Literature Several studies have evaluated the role of technologies on employee-employer matches in the labor market, including the internet (Autor, 2001) and algorithmic writing assistants like Grammarly (Wiles et al., 2025). However, to the best of our knowledge, this is the first study to provide causal evidence on how the use of LLMs by job-seekers influences signaling and matching outcomes. Unlike aforementioned technologies, LLMs can generate tailored job application materials, fundamentally changing how job-seekers present themselves to employers. As these tools become increasingly widespread, their use among job-seekers is likely to have a greater impact on the matching efficiency in the labor market.

The study most closely related to ours is Wiles et al. (2025), who randomize access to an algorithmic writing assistant (AWA) to job-seekers on an online gig platform. Unlike LLMs, the algorithmic writing assistant studied in Wiles et al. (2025) provides suggestions on text already written by the job-seeker, while LLMs can do this and generate text themselves. Importantly, our study focuses on entry-level positions in full-time employment—jobs where soft skills, often conveyed in cover letters, play a central role in selection (Deming, 2017). In contrast, the gig-economy context analyzed by Wiles et al. (2025) places less emphasis on soft skills, and involves shorter, resume-style documents with limited narrative content.^{2,3} This difference in job type and document structure implies different signaling environments and raises distinct questions about how job-seeker’s signals are interpreted by employers.

To interpret our results, we adopt the “clarity” versus “signaling” framework introduced

²Less than 1% of U.S. workers were employed in online gig roles in 2017 versus 71% in full-time jobs by 2022 (United States. Bureau of the Census and United States. Bureau of Labor Statistics, 2021; U.S. Bureau of Labor Statistics, 2023). When hiring for entry-level positions, signals are scarce because of limited past work experience, yet stakes are high due to the path dependency of future jobs (Arellano-Bover, 2024).

³Our setting differs from Wiles et al. (2025) along several other important dimensions. Firstly, the documents evaluated differ: in Wiles et al. (2025), treated job-seekers had access to AWA when writing a 70-word resume focused on their professional experience. Instead, in our study, job-seekers wrote cover letters that averaged 430 words and focused primarily on their motivation and soft skills. Secondly, the timing differs: Wiles et al. (2025) was conducted in 2021, before the emergence of LLMs, and hence recruiters took signals at face value. In contrast, our study was conducted in spring 2024, when LLM adoption among job-seekers was already widespread and recruiters were aware of this (Criddle and Strauss, 2024). Lastly, the sample differed substantially: in Wiles et al. (2025), only 20% of the sample comes from anglophone countries, with the majority of job-seekers’ first language not being English. Instead, our study consisted of highly educated individuals who had completed at least a Bachelor’s degree with English as the language of instruction. This is reflected in the spelling and grammar rate in the control group, which was 8.0% in Wiles et al. (2025), and 2.8% in our study.

by [Wiles et al. \(2025\)](#). In this framework, clarity refers to the extent by which writing reduces information frictions by making a job-seeker’s true attributes easier to assess—for example, by correcting grammatical errors or improving structure. Signaling, by contrast, captures the idea that the information conveyed in a cover letter can serve as a proxy for applicant ability: well-written applications credibly indicate competence, effort, or communication skills. In [Wiles et al. \(2025\)](#), AWAs improve clarity by reducing the number of grammatical and spelling mistakes and improving the signal’s readability, but leave the underlying signal untouched. In contrast, we find a more nuanced story: LLMs on net do not affect the clarity of the cover letter, but affect the conveyed signal. Further, LLMs improve the cover letter’s components that are not important in the recruiter’s decision to interview a candidate.

Another closely related paper is [Cowgill et al. \(2025\)](#), who estimate a slightly negative covariance between expertise and ChatGPT-induced signal gains through hiring and venture capital funding experiments, and show that generative AI compresses textual signals and reduces screening accuracy. Our results confirm a similar qualitative mechanism: access to LLMs improves the quality of cover letters among weaker job-seekers, compressing the distribution of signals and lowering recruiters’ ability to screen. Our design and setting, however, yield distinct implications. Whereas evaluators in [Cowgill et al. \(2025\)](#) are informed whether ChatGPT was used, we study screening under imperfectly observable LLM adoption. Recruiters are no better than chance at detecting LLM assistance, and disclosure interacts with recruiters’ expectations, shifting the weight they place on high-quality human-written letters. Finally, we microfound this inference problem through firm-worker complementarity and embed it in a calibrated assignment model that maps reduced signal informativeness into welfare losses via mismatch.

In particular, findings from our second experiment shed light on how recruiters form beliefs when LLM adoption is imperfectly observable, revealing that recruiters’ perceptions—thus far overlooked in the literature—play a central role in the interaction between LLM-usage and hiring decisions. While existing studies have looked at the role of AI- and LLM-generated recommendations for evaluators ([Hoffman et al., 2017](#); [Dell’Acqua, 2022](#); [Avery et al., 2024](#); [Chaturvedi and Chaturvedi, 2025](#); [Awuah et al., 2025](#)), our study looks at how evaluators perceive and judge written content created by job-seekers with and without access to LLMs. Given the widespread use of LLMs by job-seekers, this is an important distinction ([Criddle and Strauss, 2024](#)). Our second experiment also contributes to the literature on

the evaluation of human vs. LLM-generated content (Böhm et al., 2023; Noy and Zhang, 2023; Zhang and Gosline, 2023; Kadoma et al., 2024; Bohren et al., 2024) by illustrating the implications on the labor market, and by highlighting the role that perceptions of the evaluator play. Furthermore, we show that recruiters are no better than chance at correctly identifying whether a text was written with access to LLMs.

By substituting for job-seeker effort and distorting the informational content of applications, LLMs introduce a novel form of mismatch in the labor market. Unlike non-generative AI tools such as AWAs, LLMs not only reduce the discrepancy in the quality of written applications, but also blur the link between the observed signal and the job-seeker’s underlying ability. This decoupling raises important questions about how hiring decisions are made in equilibrium. To study the broader implications of these distortions, we develop an assignment model with imperfect information, calibrated using findings from our experiments. Building on the existing literatures on labor market signaling (Spence, 1973; Kurlat and Scheuer, 2020; Bassi and Nansamba, 2021; Carranza et al., 2022) and matching (Teulings, 1995; Eeckhout and Kircher, 2018), we model how LLMs alter signal quality, incorporating experimentally observed asymmetries whereby LLMs disproportionately inflate signals from lower-ability applicants. Our calibration exercise quantifies substantial welfare losses resulting from these distortions, estimating efficiency losses of up to 0.8% relative to a counterfactual scenario with no LLM adoption among job-seekers. These losses emerge because firms anticipate signal inflation, discount applications, and make suboptimal hiring decisions.

Lastly, our paper contributes to the growing body of evidence on the effects of generative AI and LLMs on productivity across a range of domains, including professional writing tasks (Noy and Zhang, 2023), law school exams (Choi and Schwarcz, 2024), coding tasks (Peng et al., 2023), consulting (Dell’Acqua et al., 2023), customer support (Brynjolfsson et al., 2025), and business practices (Otis et al., 2024). Unlike these settings, we focus on a personalized and persuasive writing task. This is an important distinction from existing studies, as LLMs may struggle with personalized persuasive writing due to their formulaic nature. Our results confirm this, as LLMs greatly improve non-personalized sections of the cover letter (e.g., introduction, closing), but do not improve the personalized sections (e.g., motivation). Textual analysis of conversations between the job-seeker and ChatGPT reveals that job-seekers repeatedly ask ChatGPT to improve the personalized sections, to no avail. In turn, ChatGPT prompts substitute google searches, resulting in fewer overall and

grammar-related searches. We also provide novel evidence that while LLMs correct spelling and grammar mistakes, they reduce readability by increasing average word length.

The remainder of this paper is structured as follows. Sections 2 and 3 discuss the experimental design and results of the job-seeker experiment, while Section 4 describes the recruiter experiment. Section 5 presents the assignment model and quantification exercise, before Section 6 concludes.

2 Job-seeker Experiment: Design

To study how LLM access affects job application signals, we conducted a field experiment with 137 students from Tilburg University and Utrecht University in collaboration with four large employers in the Netherlands: Philips, PwC, Rabobank, and VodafoneZiggo. Each firm provided an entry-level vacancy and two recruiters who evaluated applicants.

During a university career week, students enrolled in a *Cover Letter Challenge* by uploading a CV, providing basic demographics used for stratification, and selecting a preferred firm.⁴ Participants then had one hour to write a cover letter for their chosen firm. Application packages (CV and cover letter) were pseudonymized and independently evaluated by two recruiters from the relevant firm using criteria co-developed with the firms (Appendix B.1.3). Recruiters graded multiple dimensions of the CV and cover letter, assigned overall scores, and reported the likelihood of inviting the applicant to an interview on a 5-point Likert scale. Recruiters evaluated both treated and control applications but were blind to treatment status.

Participants were informed they would receive individualized feedback and that the top three cover letters (as scored by recruiters) would receive €500, €300, and €200. Recruiters could also request to be put in touch with high-scoring applicants.

We randomized participants to a control or ChatGPT-access group. The control group viewed a short placebo guide (LinkedIn job search tips) and wrote their cover letter without access to LLM websites (Appendix B.1.1). The treatment group viewed a short guide on effective prompting and was allowed to use ChatGPT (GPT-3.5) via an account pro-

⁴We recruited students in the final semester of their Bachelor’s or Master’s program who were actively searching for a job.

vided by the research team, while being subject to the same website restrictions otherwise.⁵ Browser and conversation logs indicate high compliance: 95% of treated participants used ChatGPT, while 18% of control participants attempted (unsuccessfully) to access it. A third arm (prompting training plus ChatGPT access) was pre-registered but dropped prior to implementation due to power constraints.

Randomization was stratified by university, gender, study level (BA/MA), GPA, and age. Appendix Table A1 shows balance on stratification variables. Because recruiters’ baseline CV scores are higher in the control group (Appendix Table A2), we control for CV grade in all specifications following Bruhn and McKenzie (2009); CVs were submitted prior to treatment and therefore cannot be affected by ChatGPT access.

To estimate the causal effect of ChatGPT access on cover letter evaluations and interview recommendations, we estimate:

$$Y_i = \beta_0 + \beta_1 ChatGPT_i + \gamma X_i + \mu_{f,r} + \mu_s + \varepsilon_i, \quad (1)$$

where $ChatGPT_i$ indicates assignment to the ChatGPT-access condition and X_i includes stratification variables and any imbalanced baseline covariates. We include firm-by-recruiter fixed effects ($\mu_{f,r}$) and school fixed effects (μ_s). Standard errors are clustered at the job-seeker level (Abadie et al., 2022).⁶ Estimation is by OLS unless the outcome is *Likelihood of Interview* (ordered logit) or *High Chance of Interview* (logit).

3 Job-seeker Experiment: Results

This section reports both pre-registered and exploratory results. The pre-registered outcomes are recruiter evaluations, browser history, and ChatGPT prompts (Table 1, Appendix Table A11, and Figure 2). All remaining analyses are exploratory.

3.1 Main Treatment Effects

Table 1 reports intention-to-treat estimates of ChatGPT access on recruiter-assessed cover letter quality and interview recommendations. Outcomes in columns (1)–(6) are standard-

⁵At the time of the experiment (Spring 2024), GPT-3.5 was the most advanced free version.

⁶Appendix Table A15 reports unclustered standard errors.

ized, so coefficients are interpreted in standard deviations. Access to ChatGPT increases the overall cover letter score by 0.222 standard deviations (column 1; $p < 0.10$). The improvement is concentrated in more standardized components—the introduction and closing—with effects of 0.253 and 0.281 standard deviations, respectively (columns 3 and 6; $p < 0.05$). We do not detect statistically significant effects on layout/clarity, experience, or motivation (columns 2, 4, and 5).

Table 1: Main Results: Effect of ChatGPT on Cover Letter

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
ChatGPT	0.222* (0.117)	0.161 (0.145)	0.253** (0.119)	0.075 (0.112)	0.150 (0.115)	0.281** (0.113)	0.249 (0.284)	0.011 (0.444)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000	2.799	0.278
Observations	274	274	274	274	274	274	274	274

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

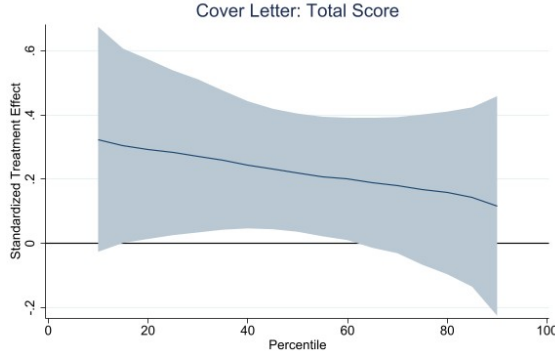
Despite higher average cover letter quality, we find no evidence that ChatGPT access increases interview recommendations. The estimated effect on the *Likelihood of Interview* measure is positive but statistically insignificant (column 7; $p = 0.380$), and the effect on the pre-registered indicator for a high interview likelihood (*Likelihood of Interview* > 3) is essentially zero (column 8; $p = 0.981$). Appendix Figure A2 shows that the implied marginal effects are small across the response categories, and Appendix Tables A7 and A8 confirm robustness to alternative inference and specification choices.⁷

To understand why improved cover letter scores do not increase interview recommendations, we examine which cover letter dimensions predict interview likelihood. Appendix Table A3 shows that the clarity and motivation components are the strongest predictors of interview recommendations. Consistent with Table 1, ChatGPT access does not measurably improve either component, providing a natural explanation for the null interview effect.

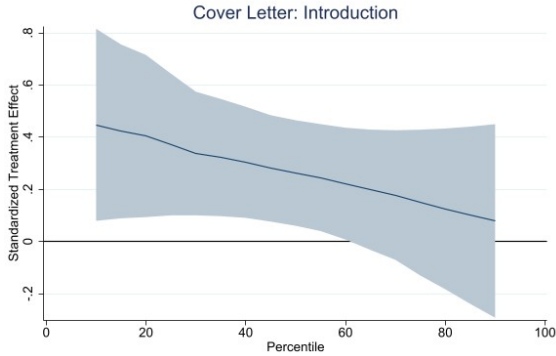
⁷We additionally verify that recruiters did not use LLMs to produce their evaluations by re-grading cover letters with ChatGPT under the same rubric (Appendix A.1.7). ChatGPT, blind to treatment status, assigns systematically higher scores to cover letters it helped generate, which is inconsistent with recruiter evaluations being produced by the same tool. This exercise also illustrates a potential evaluation bias if firms were to rely on LLM-based screening.

Figure 1. Quantile Regressions

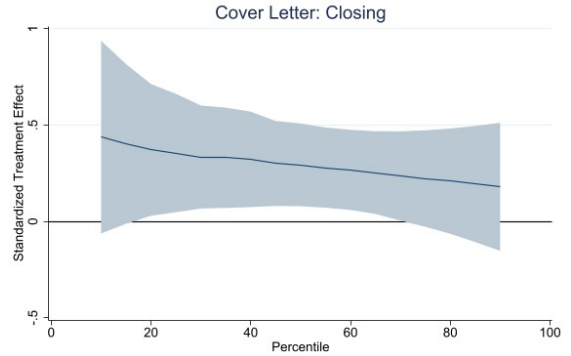
(a) Cover Letter: Total Score



(b) Cover Letter: Introduction



(c) Cover Letter: Closing



Notes: Each panel plots the estimated treatment effect of ChatGPT on cover letters across quantiles of the outcome distribution, obtained from quantile regressions. The outcomes correspond to the three main sub-components of the cover letter—Total Score, Introduction, and Closing—as described in Appendix B.1.3. Shaded areas represent 95% confidence intervals based on robust standard errors clustered at the individual level. The corresponding average (OLS) treatment effects are reported in Table 1.

We next examine heterogeneity across the distribution of applicant quality. Using quantile regressions, we find that treatment effects are concentrated among lower-scoring applicants, consistent with evidence that LLMs disproportionately benefit weaker performers (Noy and Zhang, 2023; Dell’Acqua et al., 2023; Brynjolfsson et al., 2025; Cowgill et al., 2025).⁸ Figure 1 plots quantile-specific treatment effects for the total cover letter score

⁸We find no heterogeneous treatment effects by gender or by degree type; see Appendices A.1.3 and

(Panel A), introduction (Panel B), and closing (Panel C), showing declining effects across the distribution. In contrast, heterogeneous effects for layout, motivation, and experience are not statistically significant (Appendix Figure A1).

3.2 ChatGPT’s Impact on Writing Clarity

Table 1 shows that ChatGPT access improves overall cover letter evaluations but does not affect the recruiter-rated *Layout*, *Writing Quality*, and *Clarity* dimension. To interpret this null effect, we examine objective text measures that capture two distinct dimensions of writing quality: grammatical correctness and readability (Wiles et al., 2025).

Table 2 reports treatment effects on grammatical errors, readability, and length. ChatGPT access reduces the number of grammatical errors and the error rate by 37–39% (columns 1–2; $p < 0.01$), driven primarily by fewer spelling errors (Appendix Table A17; Appendix Figures A3 and A4). However, readability deteriorates: the Flesch Reading Ease score declines by 4.45 points (about 12%; column 3; $p < 0.05$), and the Flesch–Kincaid Grade Level increases modestly (column 4; $p = 0.11$). Consistent with reduced readability, ChatGPT increases average word length by 3% (column 6; $p < 0.01$) with no detectable effect on total word count (column 5). Appendix Figures A5 and A6 show the full distributions.

Table 2: Effect of ChatGPT on Clarity and Readability

	(1)	(2)	(3)	(4)	(5)	(6)
	Grammatical Errors		Readability			Length
	Total Errors	Error Rate	Flesch Reading Ease	Flesch-Kincaid Grade Level	Word Count	Average Word Length
ChatGPT	-4.358*** (1.341)	-0.011*** (0.003)	-4.451** (1.849)	0.589 (0.373)	6.330 (22.966)	0.174*** (0.062)
Control Group Mean	11.931	0.028	35.982	13.917	428.153	6.294
Observations	137	137	137	137	137	137

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Columns 1 and 2 refer to grammatical errors, identified using the R package ‘`languageToolR`’. Columns 3 and 4 report treatment effects of the Flesch Reading Ease, and the Flesch-Kincaid Grade Level, evaluated using the R package ‘`koRpus`’. Columns 5 and 6 report treatment effects on the word count and average word length, evaluated using ChatGPT’s API. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Taken together, these results indicate that ChatGPT improves surface-level technical correctness while making the text harder to read. This helps reconcile the null effect on recruiter-rated clarity: reduced errors and reduced readability operate in opposite direc-

A.1.4. Furthermore, we document substantial variation of the correlation of within-firm recruiter evaluations, ranging from $\rho = 0.266 - 0.499$.

tions. More broadly, the divergence between technical quality and readability suggests that generative AI can alter the nature of job-application signals in ways that are not necessarily decision-relevant for screening. This distinction is consistent with the idea that LLMs may raise the stylistic sophistication of application materials without increasing their informativeness about worker quality, an implication we quantify in the matching framework in Section 5.

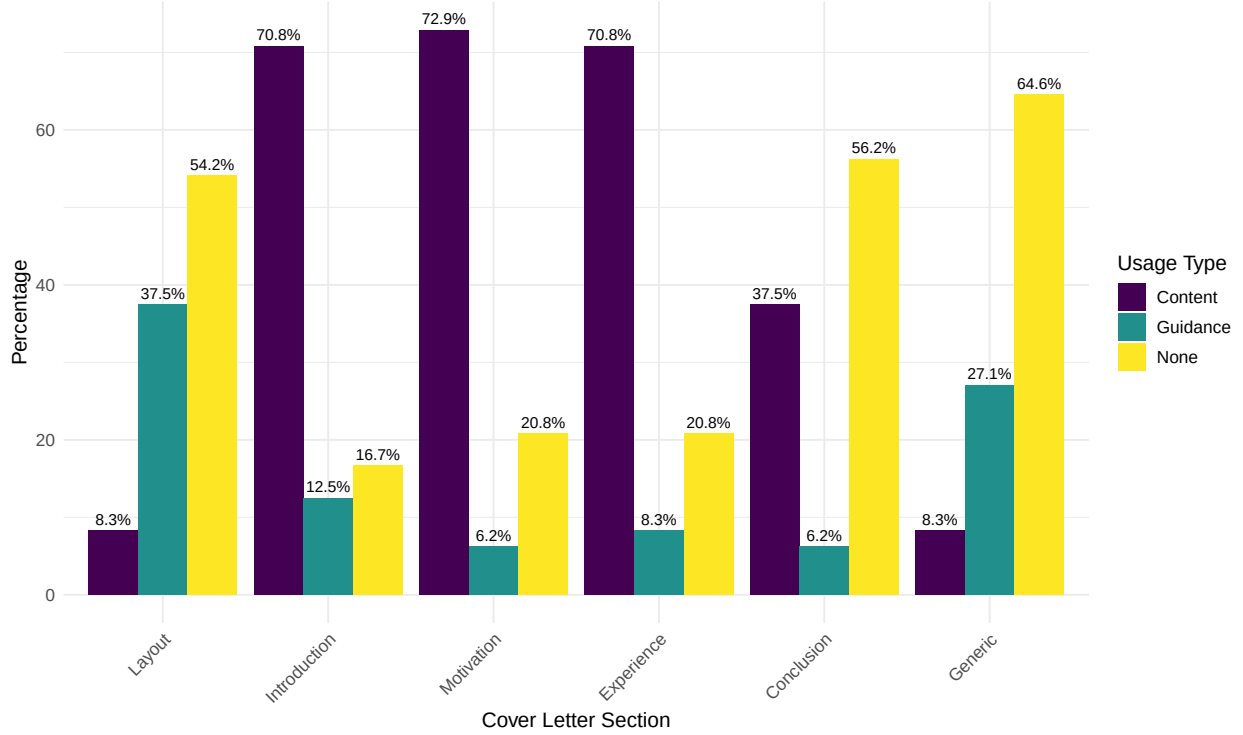
3.3 Analysis of ChatGPT Interactions

In this sub-section, we explore how job-seekers interacted with ChatGPT, how access to ChatGPT affected their browsing history, and the role of more effective prompting.

We begin by analyzing the conversations between job-seekers and ChatGPT to understand how they engaged with the LLM during the cover letter writing process. We developed a comprehensive text analysis framework that quantifies both the focus and nature of user interactions. Our methodology tracks mentions of the six key cover letter sections (Layout, Introduction, Experience, Motivation, Conclusion, and Generic, which is added to the analysis to take into account requests that users make on the entire cover letter, rather than sub-sections), distinguishing between content-based interactions (where users seek substantive help with section content) and guidance-based interactions (where users ask for simple structural or formatting advice). For each conversation, we compute the percentage of total mentions per section and the ratio of content versus guidance requests. This allows us to understand both where users focus their attention and how they utilize ChatGPT. Appendix D provides the complete technical details of our methodology, with Appendix Table A37 and Figure A14 demonstrating substantial variation in user engagement, with the number of exchanges per conversation ranging from 1 to 16 messages (mean = 6.4, median = 5).

Figure 2 shows how users target their messages across different sections during the cover letter writing process, revealing that Introduction, Motivation, and Experience sections show the highest content engagement, with approximately 71%, 73%, and 71% content requests, respectively. In contrast, Layout and Generic sections have notably different prompting patterns, with Layout showing a low share on content-based interactions (8%) and higher in guidance (38%) requests, while Generic requests are also predominantly guidance-focused (8% content, 27% guidance). Therefore, the ineffectiveness of ChatGPT at improving the

Figure 2. ChatGPT Prompts: Distribution of messages across cover letter sections.



Notes: The figure shows the distribution of ChatGPT usage across cover letter sections. Each bar reflects the share of conversations where the user requested Content (i.e., text generation or improvement), Guidance (i.e., structural or formatting advice), or showed No engagement with the section (None).

personalized sections of the cover letter is seemingly not due to job-seekers not asking for ChatGPT’s advice on these sections. Figure 2 reveals that across most sections, users predominantly engaged with ChatGPT for content-based assistance rather than guidance, suggesting applicants use ChatGPT for substantial changes to the cover letter’s content, rather than small changes in its format. The exceptions are the Layout and Generic sections, where users show greater interest in guidance rather than content-related assistance. This further underlines fundamental differences between LLMs and other technologies such as algorithmic writing assistants (Wiles et al., 2025).

Beyond section-specific engagement, we analyze four meta-categories of user behavior during cover letter writing. Figure A13 shows substantial engagement across these categories: 89% of users sought general strategy advice, 85% conducted company research, 60% made

revision requests, and 17% asked for assistance on formatting issues. The high company research engagement (85%) indicates most job-seekers used ChatGPT as an information-gathering tool beyond writing assistance, seeking insights about prospective employers, industry trends, and organizational culture. This pattern suggests ChatGPT serves as a comprehensive research platform rather than merely a text generation tool. Given the extensive use of ChatGPT for company research and strategic planning, we expect to observe corresponding reductions in traditional online information-seeking behaviors.

Our next analysis therefore investigates whether ChatGPT complements or replaces other information tools. Appendix Table A11 reveals a significant substitution effect between ChatGPT usage and other online resources: job-seekers in the treatment arm conducted 51% fewer Google searches and visited 38% fewer websites compared to individuals in the control arm (columns 1 and 2). This effect is particularly pronounced for searches related to grammar, which decrease by 84% among individuals with access to ChatGPT (column 6). These findings suggest that ChatGPT was used as a comprehensive tool, potentially replacing the need for multiple online resources. The marked reduction in grammar-related searches indicates that users may rely on ChatGPT’s language proficiency, reducing their dependence on external grammar tools, such as those evaluated in Wiles et al. (2025). This substitution effect highlights ChatGPT’s capacity to streamline the writing process by consolidating various aspects of language assistance and information gathering into a single platform.

Browser histories are not available for all job-seekers, with differential availability across experimental arms (see Appendix Table A10).⁹ Appendix Table A13 presents Lee (2009) bounds to account for potential selection bias, confirming the robustness of our findings. The consistent negative treatment effects on websites visited across these bounds further support the conclusion that ChatGPT usage substantially alters online information-seeking behavior during the cover letter writing process.

Lastly, we examine whether adherence to the ChatGPT training influenced cover letter quality, aiming to disentangle the effect of LLM usage from the guidance job-seekers received. To assess the impact of training adherence, we use OpenAI’s API to perform sentiment analysis on job-seekers’ conversation histories with ChatGPT, measuring the extent to which they applied the suggested prompting techniques. Appendix Table A14 reports

⁹When collecting browser history and ChatGPT conversations, 18 computers did not contain any recorded history.

the normalized treatment effect of compliance with the ChatGPT training on cover letter quality. The results show no significant relationship between training adherence and cover letter scores. This suggests two possibilities: either improved prompting does not meaningfully enhance cover letter quality in this context, or our training was insufficiently effective in teaching techniques that translate into better outcomes.¹⁰ These precisely estimated null results reinforce our confidence that the observed treatment effects are primarily driven by ChatGPT usage rather than the training itself. The pre-registered study design included an additional treatment arm to causally disentangle whether the treatment effect is driven by the prompting guide or access to ChatGPT, however this treatment arm was dropped due to power concerns. The correlational evidence presented in Appendix Table A14 suggests that the prompting guide did not amplify the effectiveness of the ChatGPT intervention, in line with recent studies (Bashardoust et al., 2024). Relatedly, Appendix Table A16 reports the correlational relationship between cover letter quality and number of ChatGPT prompts, with no persistent relationship found.

4 Recruiter Experiment: Perceptions of AI

The first experiment reveals that while job applicants benefit from using LLMs in their cover letters, the improvements are limited to the less critical sections. Furthermore, although access to LLMs reduced the number of spelling and grammar mistakes, it also reduced the cover letter’s readability. As a result, these enhancements do not significantly impact the likelihood of securing a job interview. However, it remains unclear if recruiters can detect LLM-generated cover letters and what their attitudes towards LLM usage are. Strong preferences could bias hiring decisions, leading to outcomes based on the recruiters’ perceptions rather than the quality of the job-seeker’s signal. This raises an important question: what happens to recruiters’ evaluations of cover letters if they are explicitly informed that some applicants used LLMs? To address this, we run a second experiment with 401 recruiters on Prolific, examining the role of LLM disclosure and its influence on recruiters’ decisions.¹¹

¹⁰The training’s content is similar to that recommended by OpenAI (see [here](#)), despite being developed independently, 10 months earlier.

¹¹Among the recruiters, 65.3% are from the United Kingdom, while the remaining 34.7% come from other European countries. All recruiters work in Human Resources, are fluent in English, and have been active on Prolific within the past 90 days.

4.1 Experimental Design

The recruiters are assigned to evaluate five pseudonymized cover letters, and are asked to evaluate each cover letter using the same grading rubric as recruiters in the job-seeker experiment.¹² The cover letters are drawn from the sample of cover letters that were written by job-seekers in the job-seeker experiment.¹³ 84% of recruiters are employed either full- or part-time, 59% identify as female, with an average age of 37. They are randomized across three treatments: *No Information*, *Partial Information*, and *Full Information*.

In the *No Information* treatment, recruiters are asked to evaluate the quality of the cover letters without receiving further information about the nature of the job-seeker experiment. Recruiters are unaware of the two experimental arms, and hence are in the same situation as the recruiters in the job-seeker experiment (and real world). In *Partial Information*, recruiters are told that some cover letters were written with the assistance of ChatGPT, while others were not. In addition to evaluating the quality of the cover letters, recruiters in this treatment are asked to identify which cover letters are written with ChatGPT’s assistance. In *Full Information*, recruiters are informed precisely which cover letters are written with the assistance of ChatGPT (akin to the setup of Cowgill et al. (2025)). This explores the scenario where applicants have to disclose LLM use in their application (which 49% of recruiters are in favor of), or if humans were able to diagnose the use of LLMs.

We run the following OLS specification to estimate the treatment effect of *Partial Information* and *Full Information* on the recruiter’s evaluation of the applicant’s cover letter and likelihood of getting interviewed, for cover letters written with and without LLMs:

¹²Recruiters are paid a fixed fee. This is in line with Carvajal et al. (2024), and several other recent studies that use unincentivized measures (Ameriks et al., 2020; Andre et al., 2022; Stango and Zinman, 2023), motivated by the literature finding limited differences between real-life behavior and related unincentivized measures (Brañas-Garza et al., 2021, 2023; Falk et al., 2023).

¹³Six cover letters are selected from both the control and treatment group in the job-seeker experiment. Cover letters are chosen such that for both treatment and control, two cover letters are from the bottom, middle, and top tercile of the total grade distribution. Furthermore, an even gender split was ensured, and all cover letters applied to the same job description. The average grades of the cover letters, as evaluated by the recruiters in the job-seeker experiment, are 5.90 vs. 5.85 for control and treatment groups ($p = 0.95$).

$$\begin{aligned}
Y_{a,r} = & \beta_0 + \beta_1 \textit{PartialInfo}_r + \beta_2 \textit{FullInfo}_r + \beta_3 \textit{GPTwritten}_r \\
& + \beta_4 \textit{PartialInfo} * \textit{GPTwritten}_r + \beta_5 \textit{FullInfo} * \textit{GPTwritten}_r \\
& + \gamma X_r + \mu_a + \varepsilon_{a,r}
\end{aligned} \tag{2}$$

where $Y_{a,r}$ is the outcome variable for the cover letter written by job applicant a , evaluated by recruiter r . $\textit{PartialInfo}_r$ and $\textit{FullInfo}_r$ are indicators equal to one if the recruiter is assigned to the *Partial Information* and *Full Information* treatment, respectively, and zero otherwise. $\textit{GPTwritten}_r$ is an indicator variable equal to one if the cover letter was written by a job-seeker who had access to LLMs, and zero otherwise. We include interactions terms between treatment assignment and the nature of the cover letter, and control for the recruiter’s age and sex (X_r) and job applicant-level fixed effects (μ_a). Following [Abadie et al. \(2022\)](#), we cluster standard errors at the recruiter level since each recruiter evaluates multiple cover letters, creating potential correlation in the error terms across observations from the same recruiter.¹⁴ Estimation is by OLS, except for when the outcome variable is *Likelihood of Interview* and *High Chance of Interview*, which are an ordered logit, and logit regression, respectively. All results presented in section 4.2 were pre-registered.

4.2 Results

Table 3 reports average treatment effects of the *Partial Information* and *Full Information* treatments on the quality of the cover letter and likelihood of inviting the candidate to an interview, for cover letters written with and without the assistance of ChatGPT. By comparing *No Information* and *Partial Information*, we observe whether priming recruiters on the potential use of ChatGPT influences their perception of the cover letter’s quality. In line with the fact that recruiters already believed that 61% of applicants use LLMs in their job applications, we find no statistically significant differences between *No Information* and *Partial Information*. Furthermore, in only 49% of cases in the *Partial Information* treatment did recruiters correctly identify whether the cover letter was written with the assistance of

¹⁴Table A30 reports treatment effects without clustering.

LLMs, statistically indistinguishable from guessing ($p = 0.5114$).¹⁵ This provides evidence that while recruiters are aware that applicants use LLMs in their job applications, they are unable to identify which applicants actually use LLMs. We furthermore do not observe a statistically significant difference in the *No Information* treatment between the evaluation of cover letters written with and without ChatGPT.

The *Full Information* treatment does not result in a higher evaluation of the overall grade of the cover letter (column 1), both for cover letters written with and without access to LLMs. Despite no difference in the evaluation of the overall quality of the cover letters, recruiters are more likely to invite job-seekers to the next stage of the recruitment process in the *Full Information* treatment when the cover letter is written without the assistance of LLMs (column 8).

Table 3: Main Results: Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.03 (0.09)	-0.10 (0.09)	-0.01 (0.09)	-0.02 (0.09)	-0.04 (0.09)	-0.05 (0.09)	-0.14 (0.15)	-0.16 (0.16)
Full	0.06 (0.09)	0.08 (0.09)	0.12 (0.09)	0.10 (0.09)	0.09 (0.09)	0.10 (0.09)	0.19 (0.16)	0.26* (0.15)
Cover Letter Written with GPT	-0.04 (0.07)	0.03 (0.07)	0.01 (0.07)	0.00 (0.06)	-0.01 (0.07)	-0.06 (0.06)	-0.10 (0.14)	-0.09 (0.14)
Partial $\times$ Written with GPT	0.04 (0.10)	0.14 (0.10)	0.03 (0.10)	0.02 (0.09)	0.05 (0.10)	0.06 (0.10)	0.14 (0.18)	0.14 (0.21)
Full $\times$ Written with GPT	0.02 (0.10)	0.02 (0.10)	-0.04 (0.09)	0.02 (0.09)	-0.07 (0.10)	-0.01 (0.09)	-0.15 (0.19)	-0.25 (0.20)
t-test Partial: ChatGPT vs. No	0.99	0.02	0.64	0.78	0.57	0.99	0.61	0.76
t-test Full: ChatGPT vs. No	0.82	0.45	0.71	0.77	0.29	0.34	0.12	0.03
t-test Partial vs. Full: No ChatGPT	0.12	0.03	0.08	0.03	0.19	0.03	0.08	0.03
t-test Partial vs. Full: Yes ChatGPT	0.31	0.41	0.40	0.11	0.92	0.28	0.82	0.80
Control Group Mean	0.00	0.00	0.00	0.00	0.00	0.00	3.40	0.53
Control Group S.D.	1.00	1.00	1.00	1.00	1.00	1.00	1.20	0.50
N	2005	2005	2005	2005	2005	2005	2005	2005

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter’s age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Comparing the *Partial Information* and *Full Information* treatments, an interesting pattern emerges: while evaluations of cover letters written with ChatGPT are unchanged

¹⁵The ability to correctly detect LLM usage is uncorrelated with the recruiter’s confidence at being able to detect LLM usage ($\rho = 0.0292, p = 0.7409$).

between the two treatments, recruiters in the *Full Information* treatment evaluated cover letters written without access to ChatGPT (which the recruiters in that treatment arm knew with certainty) statistically significantly more positively, compared to recruiters in the *Partial Information* treatment. All components of the cover letter written without access to ChatGPT—aside from the Motivation section—are evaluated more positively ($p = 0.03 - 0.08$), resulting in a statistically significantly greater likelihood of inviting the job-seeker to the next stage of the recruitment process. This indicates that recruiters are not penalizing applicants for using LLMs, but instead reward applicants that do not use them.¹⁶

Taking advantage of our experimental design, we can gain further insights into what sort of cover letters are evaluated differentially when recruiters are made aware of LLM usage. The cover letters selected from the job-seeker experiment were balanced across LLM usage vs. not, but also balanced in terms of their quality. For cover letters written both with and without the assistance of ChatGPT, two were chosen from the bottom, middle, and upper tercile, respectively. Hence we have similar variation in the quality of cover letters.

Appendix A.2.5 decomposes the average treatment effects from Table 3 for cover letters that scored in the lower and middle tercile. While cover letters in the bottom tercile written with LLM assistance scored statistically significantly better, there were no statistically significant differences across the three treatment arms. For cover letters ranked in the medium tercile, there is also no statistically significant effect of revealed LLM usage on the recruiters’ evaluation of the quality of the cover letter. Nevertheless, recruiters in the *Full Information* treatment are more likely to invite applicants with average-quality cover letters to an interview when informed that the cover letter was written without LLM assistance.

Table 4 reports treatment effects on high quality cover letters, as identified by recruiters in the job-seeker experiment. In the *No Information* treatment, recruiters evaluate cover letters written with ChatGPT assistance statistically significantly worse, also reducing the likelihood of inviting the job-seeker to an interview. This pattern is replicated in the *Partial Information* treatment, where cover letters written with ChatGPT are also evaluated statistically significantly worse. However, this does not translate into a lower likelihood of being interviewed ($p = 0.14$). In the *Full Information* treatment, the difference between cover letters written with and without LLM assistance — which the recruiters knew — was

¹⁶Appendix A.2.2 decomposes treatment effects by the gender of the recruiter, as well as their age. The positive treatment effects are primarily driven by female and older recruiters.

the most pronounced, with highly statistically significant treatment effects for the cover letter being written with LLM assistance on the overall rating of the cover letters, its individual components, and the likelihood of being invited to a job interview ($p = 0.00 - 0.05$). Therefore, recruiters reward, high-quality cover letters when they are certain that they are written independently, without LLM assistance.

Table 4: Main Results: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation - Upper Tercile

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.16 (0.12)	-0.25** (0.13)	-0.05 (0.13)	-0.08 (0.12)	-0.20* (0.12)	-0.22* (0.11)	-0.42* (0.25)	-0.56* (0.29)
Full	0.09 (0.11)	0.11 (0.12)	0.18 (0.12)	0.13 (0.11)	0.06 (0.12)	0.05 (0.11)	0.19 (0.24)	0.26 (0.31)
Cover Letter Written with GPT	-0.48*** (0.12)	-0.29** (0.13)	-0.22 (0.13)	-0.28** (0.12)	-0.48*** (0.12)	-0.69*** (0.11)	-0.92*** (0.27)	-0.90*** (0.30)
Partial $\times$ Written with GPT	0.20 (0.17)	0.29 (0.18)	0.05 (0.19)	0.03 (0.16)	0.31* (0.17)	0.34** (0.17)	0.51 (0.36)	0.54 (0.40)
Full $\times$ Written with GPT	0.04 (0.16)	0.05 (0.17)	-0.12 (0.18)	-0.03 (0.17)	0.12 (0.17)	0.11 (0.16)	-0.03 (0.35)	-0.08 (0.42)
t-test Partial: ChatGPT vs. No	0.03	0.99	0.22	0.03	0.19	0.01	0.14	0.19
t-test Full: ChatGPT vs. No	0.00	0.05	0.01	0.01	0.00	0.00	0.00	0.00
t-test Partial vs. Full: No ChatGPT	0.05	0.00	0.11	0.02	0.06	0.10	0.08	0.01
t-test Partial vs. Full: Yes ChatGPT	0.53	0.29	0.71	0.18	0.62	0.85	0.98	0.47
Control Group Mean	0.18	0.18	0.13	0.20	0.12	0.10	3.66	0.62
Control Group S.D.	0.89	0.93	0.99	0.93	0.95	0.94	1.08	0.49
N	668	668	668	668	668	668	668	668

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter’s age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of cover letters that were identified as being part of the upper tercile, based on evaluations from the first Experiment. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Comparing the *Partial Information* to the *Full Information* treatment highlights interesting conclusions too: while the recruiter’s evaluations for cover letters written with the assistance of ChatGPT do not differ statistically significantly between the two treatments, the evaluations of cover letters written without LLM assistance do. Compared to the *Partial Information* treatment, recruiters in the *Full Information* treatment arm evaluated cover letters written without LLM assistance significantly more positively when informed that the (high-quality) cover letter was written without the assistance of ChatGPT. All dimensions of the cover letter are evaluated more positively (columns 2-6), which subsequently translates into large and highly significant increases in the likelihood of inviting the candidate to a job

interview (columns 7-8).

In summary, we observe that recruiters are unable to correctly identify the use of LLMs in job applications: they perform no better than chance. However, when recruiters are informed whether job-seekers used LLMs, their evaluations of the quality of the cover letter, and likelihood to invite the applicant to a job interview, adjust. Disclosing LLM usage has no effect on the perceived quality of low- and medium-quality cover letters, but recruiters evaluate high-quality cover letters more positively when these are not written with the assistance of LLMs. This sizable increase in perceived quality also translates into a far greater likelihood of inviting the job applicant to an interview.

5 Assignment Model with Imperfect Information and LLMs

Our experiments capture the effects of LLMs at a specific moment and under an exogenous adoption rate. To assess the resulting efficiency implications, we develop an assignment model with imperfect information. Building on the literature on labor market signaling (Spence, 1973; Kurlat and Scheuer, 2020; Bassi and Nansamba, 2021; Carranza et al., 2022) and matching (Teulings, 1995; Eeckhout and Kircher, 2018), we introduce LLMs into a static one-to-one assignment framework in which firms cannot observe workers’ true abilities and instead rely on noisy signals.¹⁷ Motivated by our first experiment, we model LLMs as a signal-enhancing technology that disproportionately improves the signals of lower-ability workers, generating a non-linear distortion in the distribution of signals observed by firms. Firms form Bayesian beliefs about workers’ abilities and the likelihood that LLMs were used, and matches are determined through positive assortative matching (PAM) based on these beliefs.

¹⁷Wiles et al. (2025) also model hiring under imperfect information, but our frameworks differ along key dimensions. In Wiles et al. (2025), workers are of two types (H and L), and only H-types benefit from algorithmic writing assistance through lower effort costs or the removal of “bad writing” shocks; as a result, AWAs increase signal dispersion. In contrast, consistent with our Experiment #1, LLMs in our model disproportionately benefit lower-ability workers, compressing the distribution of signals. Moreover, firms in Wiles et al. (2025) are perfect substitutes and pay workers their expected productivity, whereas our model features heterogeneous firms and workers with positive assortative matching. This two-sided structure implies that LLM-induced signal compression generates mismatches and reduces aggregate output, while welfare effects in the partial-equilibrium model of Wiles et al. (2025) are ambiguous, as gains for high-quality workers come at the expense of low-quality ones.

While LLMs may improve signals for some workers, the resulting compression and added uncertainty reduce the informativeness of signals and distort sorting, lowering aggregate matching efficiency relative to benchmarks with either perfect information or symmetric noise without LLMs.¹⁸

We focus on PAM because it closely reflects the labor markets relevant to our empirical setting—university graduates applying to entry-level, full-time positions. PAM implies that higher-ability workers match with higher-quality firms, concentrating talent in more selective jobs, while lower-ability workers sort into less demanding roles (Arcidiacono et al., 2010). This assumption is particularly plausible for the multinational firms in our first experiment, which operate in banking, insurance, and technology and compete for graduates from selective universities.

The heterogeneous effects documented in our experiments—where LLMs disproportionately inflate signals at the bottom of the distribution—interact directly with PAM. In such markets, signal distortion obscures the true ranking of candidates and undermines efficient sorting. Lower-ability applicants with inflated signals may displace better-suited candidates at high-quality firms, while firms, anticipating this distortion, may optimally discount application materials more broadly. The model allows us to quantify these effects by comparing equilibrium outcomes with and without LLM adoption. We calibrate the model using experimental estimates of signal distortion and moments from the literature on worker and firm heterogeneity (Teulings, 1995; Eeckhout and Kircher, 2018). The calibration indicates that welfare losses relative to the no-LLM benchmark can reach up to 0.8%, highlighting the scope for misallocation in high-stakes labor markets.

A closely related distortion arises in the education literature on grade inflation. Models such as Chan et al. (2007) show that grade inflation disproportionately benefits lower-ability students, weakening the informational content of grades and deteriorating aggregate allocation. In our setting, LLMs generate an analogous distortion by compressing labor market signals. Closest to our framework, Schwager (2012) embeds grade inflation into a one-to-one

¹⁸Cowgill et al. (2025) similarly model screening with noisy signals, but treat workers’ ability and AI-induced signal gains as exogenously and jointly distributed, with an estimated slightly negative covariance implying larger gains for lower-type workers. Furthermore, AI usage is common knowledge to recruiters. In contrast, we microfound this relationship through firm–worker complementarity and allow AI use to be imperfectly observable, so that inference about AI adoption endogenously generates mismatch and welfare losses.

assignment model with imperfect information and shows that although low-ability students gain, aggregate output falls. We reach a similar conclusion: under imperfect information, the ability-dependent, non-linear distortions introduced by LLMs reduce aggregate welfare relative to a benchmark without LLM adoption.

Assignment model. The static economy consists of a continuum of workers, indexed by their quality s and distributed with CDF G_s , and a continuum of firms, indexed by their quality x and distributed with CDF G_x . We assume that both G_s and G_x admit densities denoted by g_s and g_x and have positive and bounded supports $\mathcal{S} := [\underline{s}, \bar{s}]$ and $\mathcal{X} := [\underline{x}, \bar{x}]$. Bounded supports are assumed for simplicity and do not affect our results. The value of a match between a worker of type s and a firm of type x is given by $f(x, s)$, which satisfies Assumption 1:

Assumption 1. We assume that f is increasing, concave in both arguments, and features complementarity:¹⁹ $f(x', s') - f(x, s') \geq f(x', s) - f(x, s) \forall s' \geq s \forall x' \geq x$.

Assumption 1 implies that the marginal gain from matching with a higher- x firm is larger for higher- s workers, and symmetrically that the marginal gain from a higher- s worker is larger for higher- x firms. This complementarity implies positive assortative matching: the best firms match with the best workers and the worst firms with the worst workers, as in standard production specifications in Schwager (2012) and Eeckhout and Kircher (2018).

Firms do not directly observe workers' true abilities. Instead, each worker sends a noisy signal of her ability to all firms. After observing these signals, firms form Bayesian estimates of worker quality, denoted by $\tilde{s}$, and match assortatively based on these estimates. The assignment is a function $x = \sigma(\tilde{s})$, which maps each estimated quality $\tilde{s}$ to a firm x . In equilibrium, firms and workers are matched one-to-one.²⁰ The assignment rule is therefore deterministic:

$$x = \sigma(\tilde{s}) = G_x^{-1}(G_{\tilde{s}}(\tilde{s})),$$

where $G_{\tilde{s}}$ denotes the distribution of estimated worker qualities. All randomness in matching thus comes from noise in the signal, which makes $\tilde{s}$ deviate from s . Conditional on a worker's

¹⁹This definition of complementarity is also known as weak supermodularity, and is equivalent to a non-negative cross derivative when it exists: $f_{xs} \geq 0$.

²⁰Consequently, σ must be *measure-preserving*, ensuring an equal mass of workers and firms. See Appendix E for further details on the derivation of the assignment rule.

realized signal, their match is deterministic, as in a setting where each worker sends a single application containing their signal to all firms.

We assume that signals take the following form: a fraction p of workers sends a noisy linear signal. These are workers who either do not have access to, or choose not to adopt, LLM technology. The noise has mean zero and is uncorrelated with true ability. The remaining fraction $1 - p$ adopts LLMs, which improve the quality of their signals.²¹ For a worker of true quality s , the signal y observed by firms is

$$y = \begin{cases} y_1 = s + e & \text{with probability } p, \\ y_2 = h(s + e) & \text{with probability } 1 - p, \end{cases}$$

where e is a mean-zero random variable. We impose the following assumption on h , motivated by our first experiment, where LLMs primarily improve signals for low-ability workers while leaving high-ability workers essentially unaffected:

Assumption 2. $y_2 = h(y_1)$ is a continuous function that satisfies (i) $h(y_1) \geq y_1$ for all y_1 , (ii) $0 < h'(y_1) < 1$ for all y_1 , and (iii) $\bar{y} := h(\bar{s} + \bar{e}) = \bar{s} + \bar{e}$.

Assumption 2 implies that the signal improvement is larger for low y_1 than for high y_1 , as in our job-seeker experiment (see Figure A15 for an illustration). A key feature of the information structure is that firms observe only y and do not know whether it was generated with LLM assistance. This aligns with our second experiment, where recruiters cannot reliably identify LLM usage.

Our transformation function h echoes the “grade-inflation” mapping in Schwager (2012): the incremental gain in the observed signal is largest for applicants with the lowest baseline ability. This contrasts with the clarity view in Wiles et al. (2025), in whose experiment, algorithmic writing assistants act as noise reducers, improving writing quality for every applicant and sharpening the signal without substantially altering the relative ranking of applicants. Our experimental evidence points to a different mechanism. Because the quality boost diminishes with baseline ability, h compresses the signal distribution: low-ability applicants improve substantially while high-ability applicants benefit only marginally. When LLM usage is imperfectly observable, this creates additional uncertainty about how observed text

²¹For simplicity we assume that the probability of adopting the technology is independent of s and x . In Appendix E we present an extension of the model where LLM usage is correlated with worker quality s .

maps into underlying ability. Capturing this non-linear distortion is therefore essential for assessing its consequences for sorting and welfare.

From the signal y , firms form the Bayesian estimate $\tilde{s}(y) = \mathbb{E}[s|Y = y]$ without ex ante knowing whether the signal was generated by LLMs. Firms can nonetheless infer the probability of LLM usage from the observed signal. By the law of total expectations,

$$\tilde{s}(y) = \omega(y)\mathbb{E}[s|Y_1 = y] + (1 - \omega(y))\mathbb{E}[s|Y_2 = y], \quad (3)$$

where $\omega(y)$ is the posterior probability of LLM usage conditional on y . By Bayes' rule,

$$\omega(y) = p \frac{g_{y_1}(y)}{g_y(y)} = p \frac{g_{y_1}(y)}{p g_{y_1}(y) + (1 - p) g_{y_2}(y)}.$$

If h satisfies Assumption 2, then Y_2 takes values above $\underline{y}_2 := h(\underline{s} + \underline{e})$. For signals $y < \underline{y}_2$, $g_{y_2}(y) = 0$, so $\omega(y) = 1$ and $\tilde{s}(y) = \mathbb{E}[s|Y_1 = y]$. That is, sufficiently low signals are known to come from non-users. For $y \geq \underline{y}_2$, firms use (3) to form beliefs over worker quality.

Let $\hat{s}(y) := \mathbb{E}[s|Y_1 = y]$ denote the posterior mean in an economy without LLMs, where all firms know signals come from Y_1 . Since h is monotone and invertible, $\mathbb{E}[s|Y_2 = y] = \mathbb{E}[s|Y_1 = h^{-1}(y)] = \hat{s}(h^{-1}(y))$.²² Therefore,

$$\tilde{s}(y) = \begin{cases} \hat{s}(y) & , \text{ for } y \leq \underline{y}_2, \\ \omega(y)\hat{s}(y) + (1 - \omega(y))\hat{s}(h^{-1}(y)) & , \text{ for } y \geq \underline{y}_2. \end{cases}$$

Since $h^{-1}(y) \leq y$, we have $\tilde{s}(y) < \hat{s}(y)$ for $\underline{y}_2 \leq y < \bar{y}$ (Figure 3). Thus, over the range where signals may be LLM-augmented, firms discount observed text relative to the no-LLM benchmark, making signals less informative.

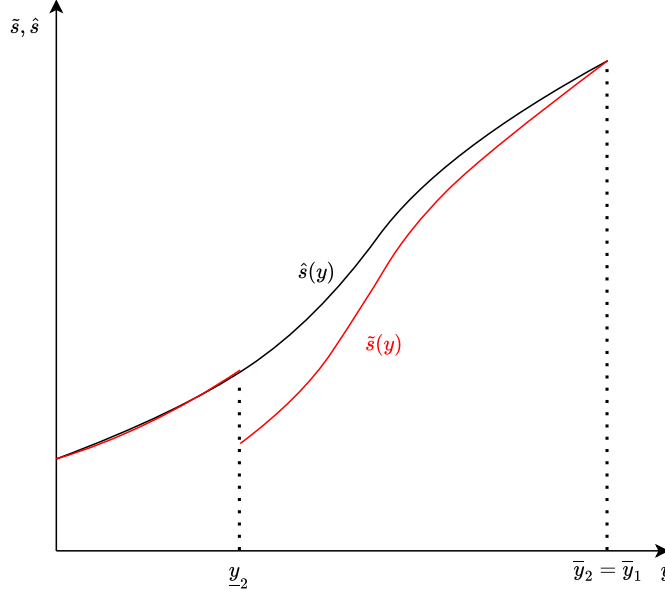
The total value in this economy with imperfect information and LLMs is

$$V_L = \int_{\mathcal{S}} \int_{\mathcal{S}} f(G_x^{-1}(G_{\tilde{s}}(\tilde{s})), s) dG_{\tilde{s}|s}(\tilde{s}|s) dG_s(s). \quad (4)$$

Proposition 1. *The total value in the economy with imperfect information and LLMs is lower than the total value of an economy with only imperfect information.*

²²See Appendix E.3 for details.

Figure 3. Comparison of $\tilde{s}$ and $\hat{s}$



Notes: The figure plots firms' expectations of worker ability without LLMs, $\hat{s}(y)$, and when LLMs are available, $\tilde{s}(y)$. Since LLMs improve signal quality, firms know that low signals must come from non-users, so the two expectations coincide for small y .

Proposition 1 states that aggregate value in the economy with imperfect information and LLMs (V_L) is at most equal to the value in the imperfect-information economy without LLMs.²³ The intuition is that while LLMs can improve some signals, imperfect observability leads firms to discount signals over the range where augmentation is possible. This induces mismatches, and with concavity and complementarity in f , lowers aggregate output relative to the no-LLM benchmark.

Calibration. To quantify the efficiency losses from LLMs under imperfect information, we calibrate the model and simulate output losses. We assume a CES production function for the value of a firm–worker match, following [Beckhout and Kircher \(2018\)](#), $f(x, s) = [\alpha x^\rho + (1 - \alpha)s^\rho]^{1/\rho}$. Table 5 reports the calibrated parameters and target moments.

Figure 4 plots efficiency losses induced by LLMs as a percentage relative to an economy with positive assortative matching and imperfect information but *without* LLMs (denoted by

²³See Appendix E for details and proof.

Table 5: Calibration Parameters

Parameter	Source	Value
Production Function	Eeckhout and Kircher (2018)	$\alpha = 0.5; \rho = 0.5$
Distribution of Worker Type	Teulings (1995)	$N(5.59, 1.29)$
Distribution of Firms Type	Teulings (1995)	truncated $N(5.59, 1.29)$
Slope of h function	Experiment #1	0.61

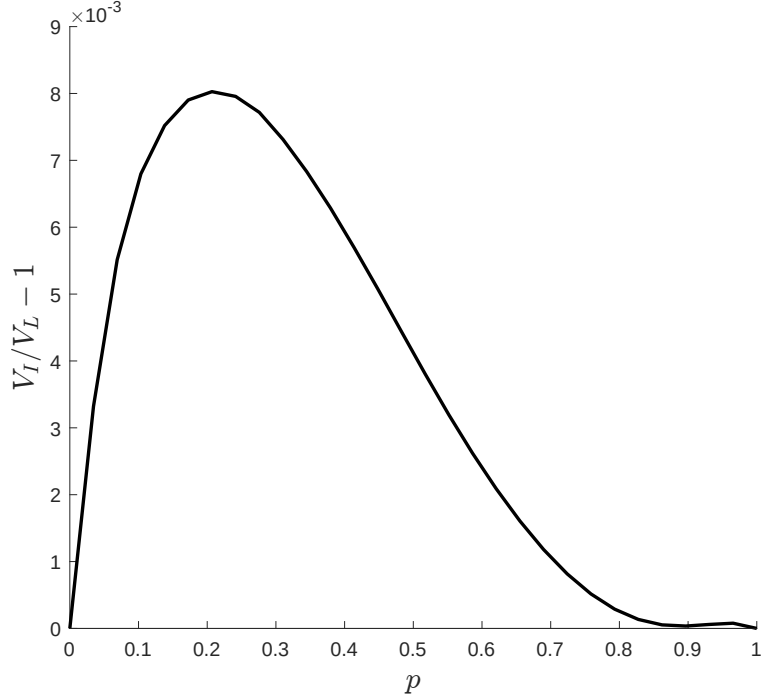
Notes: From the control group’s cover letters in Experiment #1, we have $\mathbb{E}[y_1] = 5.59$ and $Var[y_1] = 2.61$. Since $y_1 = s + e$ and e is assumed to have 0 mean, we get $E[s] = 5.59$, which we use to calibrate the moments of the workers’ type distribution. In this calibration we set the firm’s type distribution the same as the workers’ but truncated to the largest non-negative interval around the mean, i.e. $[0, 11.18]$ in order to avoid negative values. To get the slope of the h function we first calculate the quantile treatment effect (Figure 1) for each 5-th percentile, then we regress the percentiles on the treatment effects and a constant. The slope of the h function is then one minus the estimated slope coefficient, which is -0.39.

V_I). The horizontal axis reports p , the share of job-seekers who do not use LLMs. Efficiency losses peak at around 0.8% when approximately 80% of applicants use LLMs. In our job-seeker and recruiter experiments, where LLM usage rates are 47% and 50%, respectively, the implied efficiency loss is approximately 0.4%. A key feature of Figure 4 is that efficiency losses are non-monotonic in LLM adoption. When LLM usage is moderate, imperfect observability generates substantial uncertainty and misallocation. As adoption becomes nearly universal, however, this uncertainty diminishes: firms internalize that most applicants use LLMs and adjust their inference accordingly. In the limit where all applicants use LLMs ($p = 1$), firms perfectly invert the signal transformation $h(\cdot)$, so that $\tilde{s}(y) = \hat{s}(y)$ for all y , and efficiency losses disappear.

The figure also shows that LLM-induced misallocation is asymmetric around $p = 0.5$. This asymmetry reflects the distortionary effect of LLMs on signals: LLMs shift the left tail of the signal distribution upward while leaving the right tail largely unchanged, compressing signals and reducing their informativeness. As the share of LLM users increases, this compression intensifies and output falls, until adoption becomes sufficiently widespread that firms can again infer worker quality more accurately.

In Appendix E, we extend the model to allow LLM adoption to be correlated with underlying ability s . This may arise if higher-ability applicants have better access to, or are more willing to use, LLMs, or if lower-ability applicants anticipate larger gains from adoption. We model this by allowing p to vary linearly with s , holding the average adoption

Figure 4. Sensitivity of Misallocation to p



Notes: The figure shows how the total output in the economy with the presence of LLM technology V_L (calculated using equation (4)) differs from the economy where there are imperfect labor market signals but no LLM technology V_I (see Appendix E for details) when the proportion p of applicants who do *not* use LLM technology varies over $[0, 1]$.

rate fixed at 0.5. The results show that aggregate output increases as LLM adoption becomes more positively correlated with ability, consistent with the model’s mechanism: when higher-ability applicants are more likely to adopt, fewer low-ability signals are distorted, reducing mismatch and improving allocation.

6 Conclusion

The emergence of Large Language Models has the potential to significantly impact productivity and reshape labor market dynamics. LLMs are already widely adopted, with [Bick et al. \(2025\)](#) documenting that nearly 40% of the US working-age population uses them, and 23% of employees using them at least once per week. However, the influence of LLM usage by job-seekers on labor market signals and subsequent matching efficiency remains largely

unexplored. This paper addresses this gap through two field experiments involving job-seekers and recruiters, whose findings inform a calibrated assignment model with incomplete information and LLM usage.

In our job-seekers’ experiment, we find that access to OpenAI’s ChatGPT improves the average quality of cover letters. These gains, however, do not translate into higher interview probabilities, as the improvements are concentrated in less critical and less personalized sections of the cover letter. Moreover, while ChatGPT reduces spelling and grammar errors, it simultaneously lowers readability by increasing linguistic complexity. Our second experiment shows that recruiters are no better than chance at detecting LLM usage, and that disclosure interacts with recruiters’ expectations: high-quality cover letters written without LLM assistance are evaluated more positively when recruiters are aware of this, indicating a premium placed on clearly human-written applications.

Taken together, our experimental results reveal important non-linearities in how LLMs affect job-seekers. Lower-ability applicants benefit disproportionately from LLM assistance, while applicants already producing high-quality cover letters experience little improvement and may be disadvantaged when recruiters become cautious about AI-generated content. These patterns motivate our model, which embeds LLM-induced signal compression into an imperfect-information matching framework with firm-worker complementarity. The calibration shows that widespread LLM adoption can reduce aggregate welfare by distorting matching: lower-ability applicants are matched to overly demanding positions, while higher-ability applicants are displaced into less suitable roles. Although some firms or workers may benefit from this misallocation, the net effect is a decline in aggregate welfare relative to a benchmark with imperfect information but no LLM usage.

These findings are particularly relevant given the early stage of generative AI adoption, limited regulation, and uncertain future norms. If policies mandating explicit disclosure of LLM use—such as those currently discussed in the European Union—become widespread, the conditions in our *Full Information* treatment may approximate future hiring environments. More broadly, our results extend to settings beyond labor markets in which evaluators rely on written materials to assess applicants’ ability, such as university admissions.²⁴ While

²⁴Several academic journals and university admissions teams already require AI and LLM disclosure statements. The University of Michigan Law School bans AI tools in their applications, while Arizona State University Law School allows applicants to use them as long as they disclose them, and Georgia Tech offers AI guidance to applicants ([The Guardian, 2023](#)).

domains differ, the core features—written signals, imperfect observability of LLM use, and evaluator inference—are similar.

Finally, because our study relies on the freely accessible ChatGPT 3.5, our estimates likely represent a conservative lower bound on the effects of generative AI. As LLM capabilities continue to improve, understanding how these tools reshape signals, expectations, and matching remains an important avenue for future research.

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Online Appendix to:

Labor Market Signals: The Role of Large Language Models

by Kian Abbas Nejad, Giuseppe Musillo, Till Wicker, and Niccolò Zaccaria

A Figures and Tables

A.1 Job-seeker Experiment

A.1.1 Balance Table

Table A1: Balance Table: Demographics

Variable	(1) <i>Control</i>		(2) <i>Treatment</i>		(1)-(2) Pairwise t-test
	N	Mean/(SD)	N	Mean/(SD)	P-value
Dutch Speaker	72	0.236 (0.428)	65	0.200 (0.403)	0.613
GPA	72	7.505 (0.669)	65	7.557 (0.639)	0.644
English Ability	72	6.069 (0.738)	65	6.138 (0.788)	0.597
Master's Student	72	0.528 (0.503)	65	0.585 (0.497)	0.507
Age	72	23.597 (4.023)	65	23.123 (3.347)	0.457
Econ or Business Degree	72	0.750 (0.436)	65	0.738 (0.443)	0.878
Tilburg University	72	0.722 (0.451)	65	0.677 (0.471)	0.567
Female	72	0.486 (0.503)	65	0.538 (0.502)	0.544

Notes: Dutch speaker is a dummy variable equal to one if the student self-reports that they speak Dutch. GPA is on a scale from 0-10, and self-reported. English ability is on a scale from 1-7. Master's student is a dummy variable if the student is enrolled in a Master's program. Age is the student's age. Econ or Business Degree is a dummy variable if the student is enrolled at Tilburg's School of Economics and Management, or Utrecht's School of Economics. Tilburg University is a dummy equal to one if the student is enrolled at Tilburg University. Female is a dummy equal to one if the student identifies as a female.

Table A2: Balance Table: CV Evaluation

Variable	(1) <i>Control</i>		(2) <i>Treatment</i>		(1)-(2) Pairwise t-test
	N	Mean/(SD)	N	Mean/(SD)	P-value
CV Grade	144	5.777 (1.567)	130	5.428 (1.663)	0.075*
CV: Layout	144	0.017 (0.997)	130	-0.019 (1.007)	0.768
CV: Education	144	0.107 (1.000)	130	-0.119 (0.991)	0.062*
CV: Experience	144	0.102 (0.960)	130	-0.113 (1.034)	0.077*
CV: Extra Curricular	144	0.081 (0.944)	130	-0.089 (1.055)	0.161

Notes: The sub-components of the CV are described in more detail in Appendix B.1.3. Evaluations of the sections are on a scale from 0-10.

A.1.2 Determinants of Interview Likelihood

Table A3: Determinants of Interview Likelihood

	(1) Likelihood of Interview
CL: Layout	0.525** (0.216)
CL: Intro	0.293 (0.249)
CL: Experience	0.399 (0.249)
CL: Motivation	0.823*** (0.249)
CL: Closing	0.474 (0.315)
Control Group Mean	2.799
Observations	274

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Ordered Logit regression, regressing the likelihood of an interview to a job interview, on the sub-components of the cover letter and CV (described in Appendix B.1.3). Sub-components of the CV are omitted from the regression table, for visibility purposes. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.3 Heterogeneity - By Gender

Table A4: Heterogeneous Results: Men vs. Women

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
ChatGPT	0.155 (0.176)	0.256 (0.208)	0.196 (0.191)	0.059 (0.153)	-0.030 (0.185)	0.211 (0.193)	0.194 (0.354)	0.047 (0.522)
Female	-0.022 (0.170)	0.109 (0.180)	0.001 (0.177)	-0.026 (0.170)	-0.205 (0.159)	0.078 (0.197)	0.127 (0.414)	0.207 (0.617)
ChatGPT*Female	0.123 (0.236)	-0.215 (0.276)	0.072 (0.239)	0.058 (0.222)	0.335 (0.237)	0.170 (0.244)	0.035 (0.577)	-0.087 (0.875)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000	2.799	0.278
Observations	274	274	274	274	274	274	274	274

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Female is a dummy variable equal to one if the job-seeker identifies as a female, and zero otherwise. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.4 Heterogeneity - By Economics Degree vs. Not

Table A5: Heterogeneous Results: Econ vs. Non-Econ degree

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
ChatGPT	0.267 (0.247)	0.360 (0.259)	0.152 (0.197)	-0.085 (0.296)	0.238 (0.267)	0.408 (0.256)	0.083 (0.669)	-0.298 (0.886)
Econ/Business Degree	0.078 (0.221)	-0.067 (0.220)	-0.033 (0.191)	-0.077 (0.267)	0.229 (0.220)	0.231 (0.230)	0.904* (0.508)	0.861 (0.785)
ChatGPT*Econ/Business	-0.066 (0.282)	-0.279 (0.303)	0.107 (0.252)	0.228 (0.323)	-0.130 (0.292)	-0.150 (0.287)	0.128 (0.724)	0.451 (1.009)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000	2.799	0.278
Observations	274	274	274	274	274	274	274	274

Standard errors in parentheses

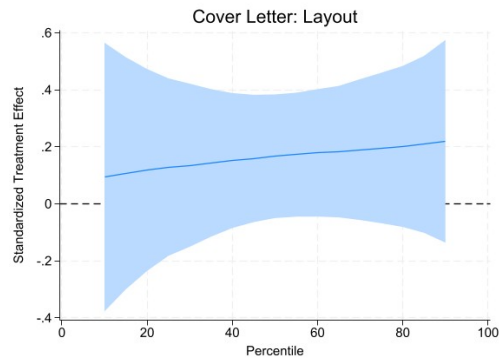
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Econ/Business is a dummy variable equal to 1 if students are enrolled in Tilburg University's School of Economics and Management, or the Utrecht School of Economics. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

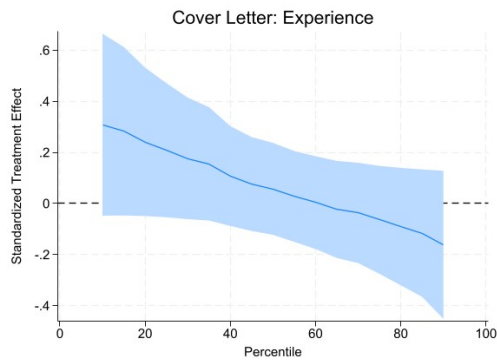
A.1.5 Heterogeneity by Ability: Layout, Motivation, Experience

Figure A1. Quantile Regressions

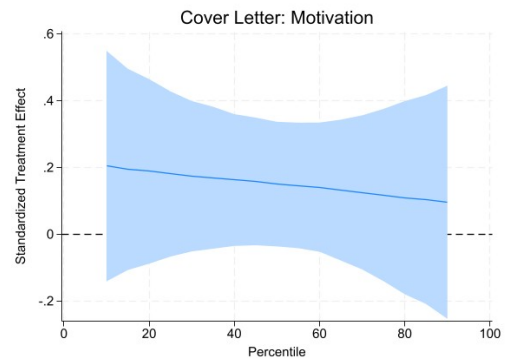
(a) Cover Letter: Layout



(b) Cover Letter: Motivation



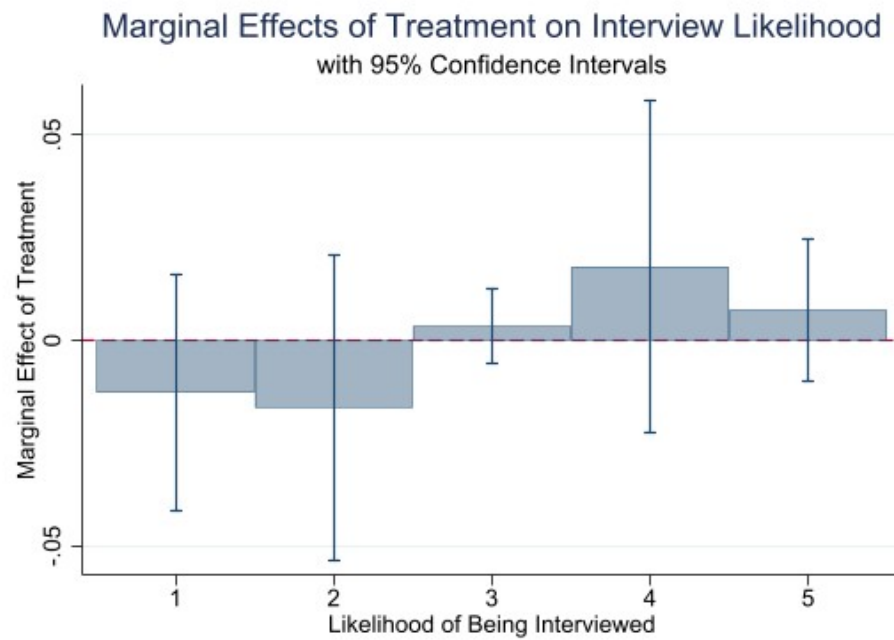
(c) Cover Letter: Experience



Notes: Each panel plots the estimated treatment effect of ChatGPT on cover letters across quantiles of the outcome distribution, obtained from quantile regressions. The outcomes correspond to the three main sub-components of the cover letter—Layout, Motivation, and Experience—as described in Appendix B.1.3. Shaded areas represent 95% confidence intervals based on robust standard errors clustered at the individual level. The corresponding average (OLS) treatment effects are reported in Table 1.

A.1.6 Marginal Treatment Effects: Likelihood of Interview

Figure A2. Marginal Treatment Effects: Likelihood of Interview



Notes: The figure shows the marginal treatment effect of being assigned to the treatment arm in Experiment #1, on the likelihood of receiving a score from 1 to 5 on the likelihood of being interviewed (higher scores are better).

A.1.7 ChatGPT as a Recruiter

Table A6: ChatGPT as the Recruiter

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	ChatGPT as the Recruiter							
	Total	Layout	Cover Letter		Motivation	Closing	Likelihood of Interview	High Chance of Interview
			Intro	Experience				
ChatGPT	0.286*** (0.099)	0.219*** (0.074)	0.264** (0.133)	0.213* (0.114)	0.286** (0.116)	0.377*** (0.119)	0.981* (0.530)	2.218 (2.565)
Control Group Mean	0.000	0.000	0.000	0.000	-0.000	0.000	4.028	0.875
Observations	137	137	137	137	137	137	137	128

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Column 8 contains 9 fewer observations because one of the 'school fixed effect' dummies perfectly predicts the outcome variable, and hence those observations are dropped. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.8 Robustness: Double Lasso (Belloni et al., 2014)

Table A7: Double Lasso regression

	(1)	(2)	(3)	(4)	(5)	(6)
	Cover Letter					
	Total	Layout	Intro	Experience	Motivation	Closing
ChatGPT	0.218** (0.111)	0.170 (0.138)	0.223** (0.114)	0.094 (0.105)	0.129 (0.112)	0.296** (0.117)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000
Observations	274	274	274	274	274	274

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions. All regressions use the PD Lasso technique to identify controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-6 refer to variables as described in Appendix B.1.3. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.9 Robustness: Bootstrap

Table A8: Bootstrap (200 reps)

	(1)	(2)	(3)	(4)	(5)	(6)
	Cover Letter					
	Total	Layout	Intro	Experience	Motivation	Closing
ChatGPT	0.222* (0.118)	0.161 (0.144)	0.253** (0.121)	0.075 (0.108)	0.150 (0.120)	0.281** (0.115)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000
Observations	274	274	274	274	274	274

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level, and bootstrapped with 200 repetitions. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-6 refer to variables as described in Appendix B.1.3. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.10 Time Taken

Another dimension along which job-seekers using ChatGPT and not using ChatGPT could differ is the time taken to write the cover letter. If ChatGPT substitutes ones own writing, job-seekers can write cover letters more quickly. On the contrary, if job-seekers use ChatGPT to complement their own writing, the whole process can take longer. Table A9 reports average treatment effects for the time taken to write cover letters, with the use of ChatGPT reducing the amount of time job-seekers spent writing a cover letter by 2 minutes on average, a result which is not statistically significant. This null result is robust to winsorizing individuals that wrote for more than 60 and 75 minutes, the recommended time job-seekers had.

Table A9: Time Taken to Write Cover Letter

	(1)	(2)	(3)
	Time Taken		
	No wins.	Wins. 75 min	Wins. 60 min
ChatGPT	-2.120 (1.815)	-1.568 (1.679)	-0.327 (1.422)
Control Group Mean	57.13	56.62	54.88
Observations	132	132	132

Notes: Intention to Treat estimates from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Time taken is the amount of time students took to write the cover letter. Columns 2 and 3 winsorize the time taken at 75 and 60 minutes, respectively. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.11 Browser History

Balance Table: Those with vs. without browser history

Table A10: Balance Table: Browser History

Variable	(1)		(2)		(1)-(2) Pairwise t-test P-value
	<i>Has Browser History</i>	Mean/(SD)	<i>No Browser History</i>	Mean/(SD)	
Cover Letter Grade	119	-0.049 (1.070)	18	0.083 (0.960)	0.621
CV Grade	119	5.636 (1.670)	18	5.500 (1.727)	0.749
Assignment to Treatment	119	0.521 (0.502)	18	0.167 (0.383)	0.005***
Speaks Dutch	119	0.202 (0.403)	18	0.333 (0.485)	0.211
GPA	119	7.570 (0.640)	18	7.263 (0.695)	0.063*
English Proficiency	119	6.118 (0.750)	18	6.000 (0.840)	0.542
Master's Student	119	1.546 (0.500)	18	1.611 (0.502)	0.609
Age	119	23.101 (3.583)	18	25.167 (4.148)	0.027**
Econ/Business Degree	119	0.748 (0.436)	18	0.722 (0.461)	0.818
At Tilburg University	119	0.689 (0.465)	18	0.778 (0.428)	0.447
Female	119	0.521 (0.502)	18	0.444 (0.511)	0.548

Notes: Cover Letter and CV Grade are the average cover letter and CV grades, as evaluated by the recruiters. Assignment to Treatment is a dummy equal to one if the student was assigned to the ChatGPT treatment. Dutch speaker is a dummy variable equal to one if the student self-reports that they speak Dutch. GPA is on a scale from 0-10, and self-reported. English ability is on a scale from 1-7. Master's student is a dummy variable if the student is enrolled in a Master's program. Age is the student's age. Econ or Business Degree is a dummy variable if the student is enrolled at Tilburg's School of Economics and Management, or Utrecht's School of Economics. Tilburg University is a dummy equal to one if the student is enrolled at Tilburg University. Female is a dummy equal to one if the student identifies as a female.

Table A11: Treatment Effects: Browser History

	(1)	(2)	(3)	(4)	(5)	(6)
	# Websites Visited	# Google Searches	Searched: Firm	Searched: Cover Letter	Searched: Grammar	Searched: Translation
ChatGPT	-7.422*** (2.215)	-5.120*** (1.320)	-1.825 (1.169)	-1.140 (0.692)	-2.427*** (0.631)	-1.280 (0.749)
Control Group Mean	19.351	10.000	5.281	2.368	2.895	2.175
Observations	119	119	119	119	119	119

Notes: Intention to Treat estimates. Column 1 reports treatment effects on the total number of websites visited, while Column 2 refers to the number of google searches. Columns 3-6 refer to whether the subject searched the relevant topic. Treatment effects are reported from OLS regressions. Standard errors are clustered at the individual level. PD Lasso machine learning technique is used to select control variables (Belloni et al., 2014), along with firm and school-level fixed effects. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. The sample consists of job-seekers in the Control and Treatment arms, for whom browser histories were saved. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Lee Bounds (2009) Calculation

Table A12: Probability of Not Having Browser History

	(1) No Browser History
ChatGPT	-0.137** (0.053)
Control Group Mean	0.208
Observations	137

Notes: Intention to Treat estimates from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. No Browser History is a dummy variable equal to 1 if we were unable to obtain the browser history of the student, and zero otherwise. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

There is differential ‘attrition’ of the browser history across treatments. In the Control group, we have the browser history for 79.17% of the sample. In the Treatment group, we have the browser history for 95.38% of the sample. The differential attrition rate is $95.38\% - 79.17\% = 16.22\%$. This is equal to $16.22\% / 95.38\% = 17.00\%$ of the Control Group sample.

To get a lower bound, we trim the top 17.00% of the control group (for each outcome variable). To get an upper bound, we trim the bottom 17.00% of the control group (for each outcome variable).

Table A13: Browser History

	(1)	(2)	(3)	(4)	(5)	(6)
	# Websites Visited	# Google Searches	Searched: Firm	Searched: Cover Letter	Searched: Grammar	Searched: Translation
<i>Panel A. Lee (2009) Upper Bound</i>						
ChatGPT	-2.220 (1.425)	-1.939* (0.906)	1.327* (0.552)	0.545 (0.363)	-0.958** (0.317)	0.486 (0.268)
Control Group Mean	19.351	10.000	5.281	2.368	2.895	2.175
Observations	110	110	111	111	112	110
<i>Panel B. Lee (2009) Lower Bound</i>						
ChatGPT	-9.484*** (2.247)	-6.341*** (1.365)	-1.825 (1.169)	-1.140 (0.692)	-2.427*** (0.631)	-1.280 (0.749)
Control Group Mean	19.351	10.000	5.281	2.368	2.895	2.175
Observations	112	111	119	119	119	119

Notes: Intention to Treat estimates. Column 1 reports treatment treatment effects on the total number of websites visited, while Column 2 refers to the number of google searches. Columns 3-6 refer to whether the subject searched the relevant topic. Treatment effects are reported from OLS regressions. Standard errors are clustered at the individual level. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Panels A and B correct for Lee (2009) bounds, as explained in Appendix A.1.11. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.12 Training and Treatment Effects

Table A14: Effect of ChatGPT Training on Results

	(1)	(2)	(3)	(4)	(5)	(6)
	Cover Letter					
	Total	Layout	Intro	Experience	Motivation	Closing
ChatGPT Training Compliance	-0.001 (0.100)	0.021 (0.115)	0.046 (0.084)	-0.111 (0.070)	-0.010 (0.099)	0.046 (0.107)
Control Group Mean	-0.056	-0.051	-0.090	0.009	-0.021	-0.094
Observations	82	82	82	82	82	82

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Columns 1-6 refer to variables as described in Appendix B.1.3. ChatGPT training compliance is a 10-point scale indicating whether individuals applied the ten techniques discussed for effective prompting, see Appendix B.1.2. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively. The sample consists of job-seekers in the Treatment arm, for whom ChatGPT conversation histories were saved.

A.1.13 No Clustered Standard Errors

Table A15: Main Results: Effect of ChatGPT on Cover Letter

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Cover Letter						Likelihood of	High Chance of
	Total	Layout	Intro	Experience	Motivation	Closing	Interview	Interview
ChatGPT	0.222** (0.095)	0.161 (0.116)	0.253** (0.099)	0.075 (0.098)	0.150 (0.100)	0.281*** (0.092)	0.249 (0.259)	0.011 (0.414)
Control Group Mean	0.000	0.000	0.000	0.000	0.000	0.000	2.799	0.278
Observations	274	274	274	274	274	274	274	274

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are robust. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.1.14 Correlation between Prompts and Cover Letter Quality

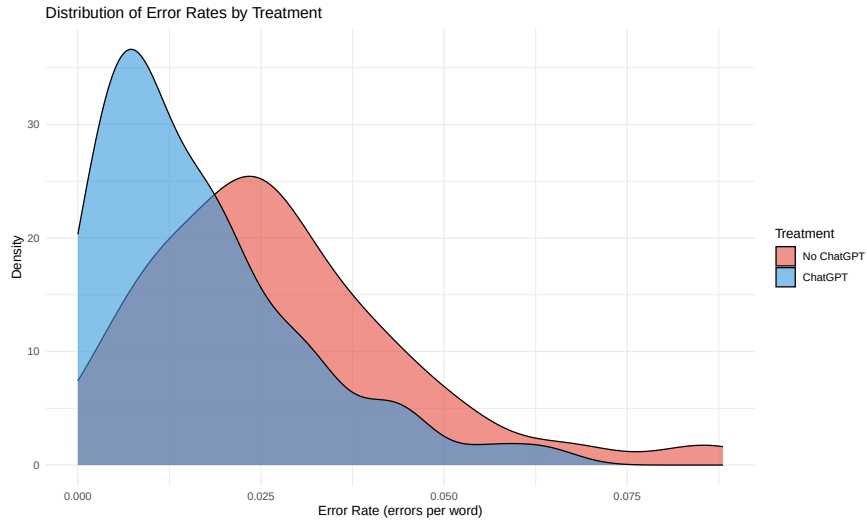
Table A16: OLS Regression: ChatGPT Prompts and Cover Letter Quality

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Number of ChatGPT Prompts	0.016 (0.018)	0.017 (0.026)	0.022 (0.016)	-0.023 (0.016)	0.016 (0.022)	0.032* (0.018)	0.008 (0.071)	0.094 (0.104)
Observations	94	94	94	94	94	94	94	94

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Observations refer to individuals assigned to the treatment group, for whom LLM conversation histories are available. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

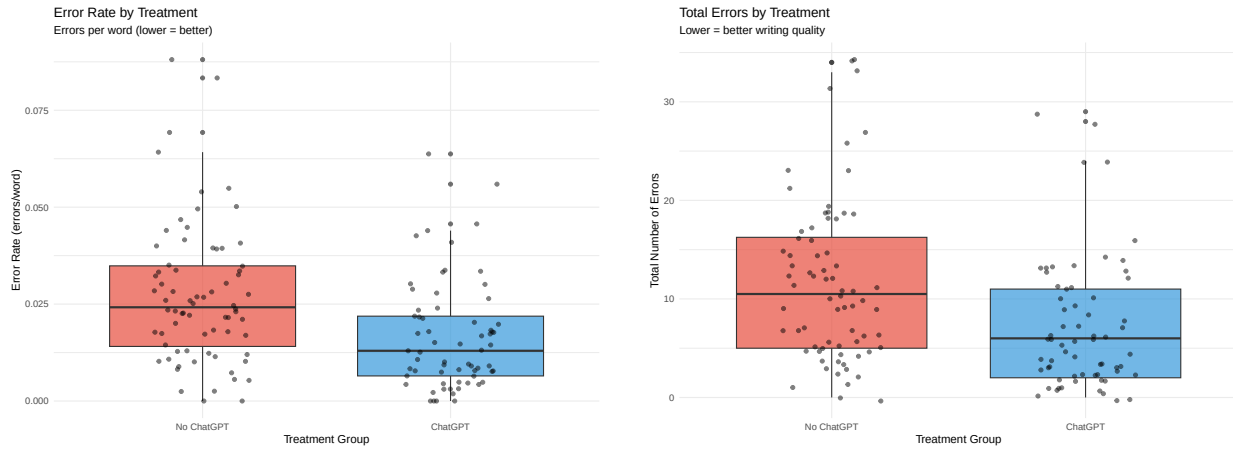
A.1.15 Readability and Error Analysis

Figure A3. Distribution of Error Rates by Treatment



Notes: The figure shows the distribution of overall error rates (errors per word) by treatment group. ChatGPT systematically shifts the distribution toward lower error rates.

Figure A4. Technical Writing Quality by Treatment Group



(a) Error Rate by Treatment

(b) Total Errors by Treatment

Notes: The figure shows box plots of writing error metrics by treatment group. Error rate is calculated as total errors divided by word count. ChatGPT significantly reduces both error rates and total error counts across cover letters, with medium effect sizes (Cohen's $d > 0.6$).

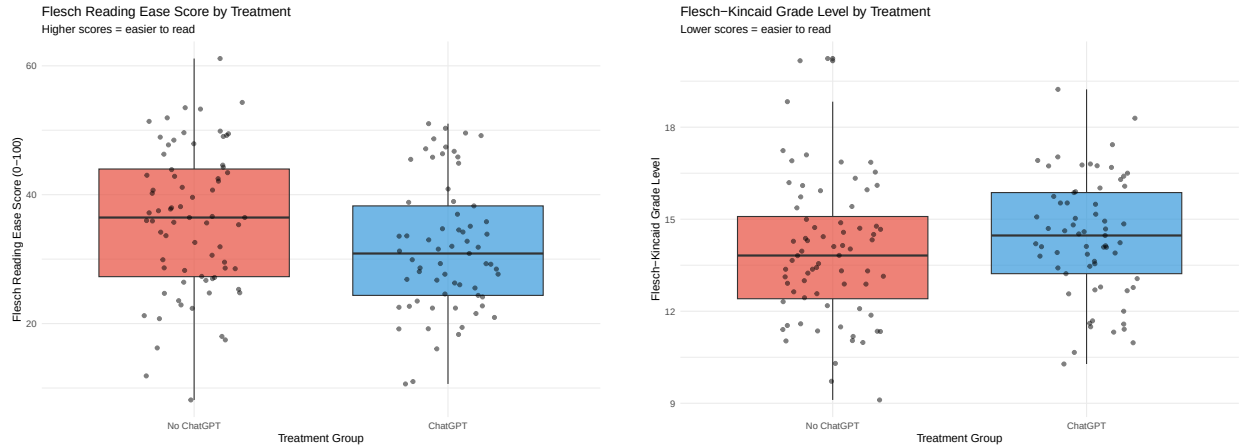
Table A17: Effect of ChatGPT on Technical Writing Errors

	(1)	(2)	(3)	(4)	(5)	(6)
	Capitalization	Spelling	Grammar	Punctuation	Typography	Style
ChatGPT	0.049 (0.045)	-3.406*** (1.044)	-0.290 (0.206)	-0.236 (0.193)	0.000 (.)	-0.304** (0.138)
Control Group Mean	0.028	9.000	0.806	0.681	0.000	0.625
Observations	137	137	137	137	137	137

	(7)	(8)	(9)	(10)	(11)	(12)
	Misc.	Redundant Phrases	Non-standard Phrases	Commonly Confused Words	Collocations	Semantics
ChatGPT	-0.221 (0.280)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.050 (0.053)
Control Group Mean	0.722	0.000	0.000	0.000	0.000	0.069
Observations	137	137	137	137	137	137

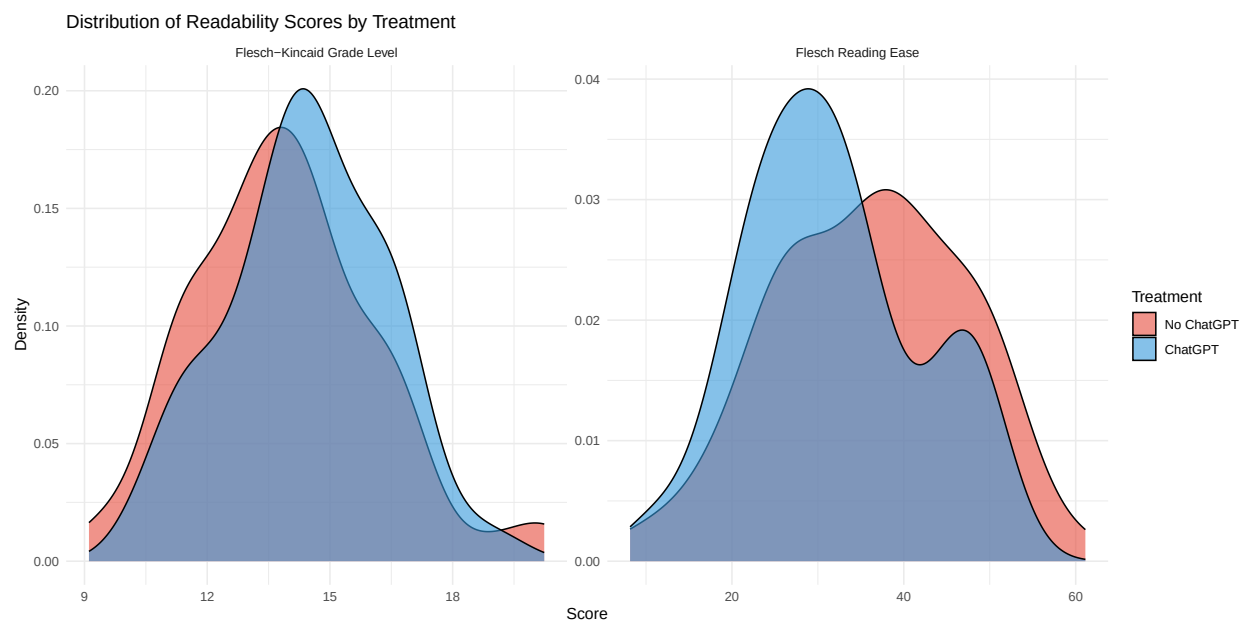
Notes:

Notes: Intention to Treat estimates from OLS regressions. All regressions include strata variables and imbalanced baseline variables as controls, along with recruiter and school-level fixed effects. Standard errors are clustered at the individual level. Control mean refers to the mean value of the outcome in the control group. Robust standard errors are in parentheses. Error rule categories are based on the R package ‘`languageToolR`’, and explained thoroughly in [Wiles et al. \(2025\)](#) Table A4. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Figure A6. Readability Metrics by Treatment Group**(a)** Flesch Reading Ease**(b)** Flesch-Kincaid Grade Level

Notes: The figure shows readability scores by treatment group. Higher Flesch Reading Ease scores indicate easier-to-read text, while higher Flesch-Kincaid Grade Levels indicate more complex text requiring higher education levels to understand. ChatGPT significantly reduces reading ease while increasing grade-level complexity.

Figure A5. Distribution of Readability Scores by Treatment

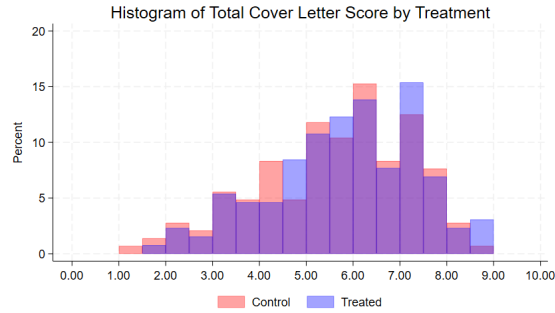


Notes: The figure shows the distribution of readability metrics by treatment group. Panel A shows Flesch Reading Ease scores (higher = easier to read). Panel B shows Flesch-Kincaid Grade Levels (higher = more complex). ChatGPT systematically shifts the distribution toward lower readability despite improving technical quality.

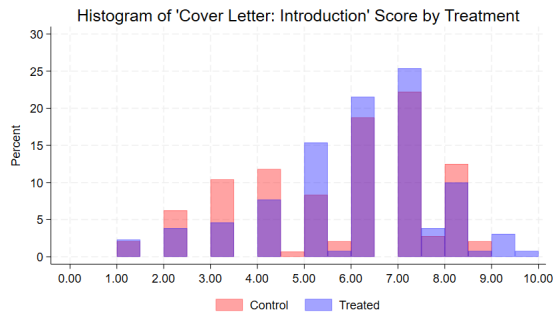
A.1.16 Histograms of Cover Letter Raw Mean Grades

Figure A7. Raw Grades, Overlapping Histograms per Treatment

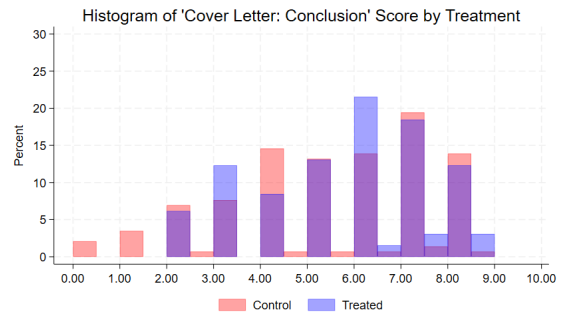
(a) Cover Letter: Total Score



(b) Cover Letter: Introduction



(c) Cover Letter: Closing



Notes: The figures show histograms of the recruiter evaluations of the job-seekers' cover letters. In particular, (a) reports the final grade; (b) reports the grade for the introduction; and (c) reports the grade for the conclusion. Blue bars reflect grades from job-seekers assigned to the treatment arm, while red bars reflect grades from job-seekers assigned to the control arm of Experiment #1.

A.2 Recruiter Experiment

A.2.1 Balance Table

Table A18: Balance Table: Demographics and Duration

Variable	(1) <i>No Info</i>		(2) <i>Partial Info</i>		(3) <i>Full Info</i>		(1)-(2)	(1)-(3)	(2)-(3)
	N	Mean/(SD)	N	Mean/(SD)	N	Mean/(SD)	Pairwise t-test P-value		
Duration (mins)	132	20.290 (10.066)	132	19.230 (10.733)	137	20.224 (10.736)	0.409	0.959	0.448
Intro Duration (mins)	132	3.202 (3.312)	132	3.120 (3.932)	137	3.297 (4.013)	0.854	0.833	0.715
Age	132	36.879 (11.072)	132	37.697 (11.499)	137	35.911 (11.324)	0.556	0.479	0.200
Female	132	0.598 (0.483)	132	0.598 (0.492)	137	0.562 (0.498)	1.000	0.547	0.547
Full-/Part-time	132	0.826 (0.381)	132	0.833 (0.374)	137	0.861 (0.347)	0.871	0.424	0.525
Student	132	0.250 (0.435)	132	0.174 (0.381)	137	0.248 (0.434)	0.133	0.973	0.139

Notes: Duration is the total number of minutes that the recruiter spent on the experiment, while Intro Duration captures the duration in minutes that the recruiter spent on the introduction. Age is the recruiter's age. Female is a dummy equal to one if the student identifies as a female. Full-/Part-time is a dummy variable equal to one if the recruiter is employed in full- or part-time employment. Student is a dummy equal to one if the recruiter is a student.

A.2.2 Additional Results

Table A19: Additional Results: Full Sample

	(1)	(2)
	Want to See CV	Time Taken
Partial	-0.07 (0.17)	-0.01 (0.11)
Full	-0.01 (0.19)	-0.01 (0.10)
Cover Letter Written with GPT	0.93*** (0.24)	0.00 (.)
Partial $\times$ Written with GPT	-0.38** (0.17)	-0.05 (0.11)
Full $\times$ Written with GPT	-0.12 (0.18)	-0.03 (0.11)
t-test Partial: ChatGPT vs. No	0.75	0.27
t-test Full: ChatGPT vs. No	0.17	0.27
t-test Partial vs. Full: No ChatGPT	0.10	0.84
t-test Partial vs. Full: Yes ChatGPT	0.04	0.80
Control Group Mean	4.46	-0.00
Control Group S.D.	1.41	1.00
Observations	2005	2005

Notes: Intention to Treat estimates from OLS regressions. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses. Column (1) reports the recruiter's need to see the CV (on a 7-point Likert scale), while Column (2) reports the normalized time taken by the recruiter to read and evaluate the cover letter. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A20: Additional Results: Upper Tercile Cover Letters

	(1)	(2)
	Want to See CV	Time Taken
Partial	-0.18 (0.24)	-0.17 (0.14)
Full	0.11 (0.26)	-0.03 (0.15)
Cover Letter Written with GPT	0.69*** (0.25)	0.00 (.)
Partial $\times$ Written with GPT	-0.14 (0.32)	-0.02 (0.17)
Full $\times$ Written with GPT	0.03 (0.33)	0.01 (0.18)
t-test Partial: ChatGPT vs. No	0.29	0.45
t-test Full: ChatGPT vs. No	0.21	0.70
t-test Partial vs. Full: No ChatGPT	0.05	0.05
t-test Partial vs. Full: Yes ChatGPT	0.13	0.12
Control Group Mean	4.54	-0.00
Control Group S.D.	1.26	1.00
Observations	668	668

Notes: Intention to Treat estimates from OLS regressions. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses. Column (1) reports the recruiter's need to see the CV (on a 7-point Likert scale), while Column (2) reports the normalized time taken by the recruiter to read and evaluate the cover letter. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A21: Additional Results: Medium-Tercile Cover Letters

	(1)	(2)
	Want to See CV	Time Taken
Partial	-0.33 (0.23)	0.11 (0.16)
Full	-0.37 (0.26)	-0.08 (0.12)
Cover Letter Written with GPT	-0.45 (0.28)	0.00 (.)
Partial $\times$ Written with GPT	-0.03 (0.29)	-0.25 (0.19)
Full $\times$ Written with GPT	0.34 (0.32)	0.03 (0.16)
t-test Partial: ChatGPT vs. No	0.23	0.23
t-test Full: ChatGPT vs. No	0.58	0.40
t-test Partial vs. Full: No ChatGPT	0.41	0.62
t-test Partial vs. Full: Yes ChatGPT	0.17	0.49
Control Group Mean	4.55	-0.00
Control Group S.D.	1.33	1.00
Observations	668	668

Notes: Intention to Treat estimates from OLS regressions. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses. Column (1) reports the recruiter's need to see the CV (on a 7-point Likert scale), while Column (2) reports the normalized time taken by the recruiter to read and evaluate the cover letter. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A22: Additional Results: Low-Tercile Cover Letters

	(1) Want to See CV	(2) Time Taken
Partial	0.31 (0.28)	0.09 (0.20)
Full	0.22 (0.28)	0.12 (0.16)
Cover Letter Written with GPT	0.51** (0.26)	0.00 (.)
Partial $\times$ Written with GPT	-0.94*** (0.30)	0.12 (0.27)
Full $\times$ Written with GPT	-0.67** (0.30)	-0.18 (0.19)
t-test Partial: ChatGPT vs. No	0.70	0.82
t-test Full: ChatGPT vs. No	0.52	0.02
t-test Partial vs. Full: No ChatGPT	0.85	0.40
t-test Partial vs. Full: Yes ChatGPT	0.48	0.16
Control Group Mean	4.29	0.00
Control Group S.D.	1.60	1.00
Observations	669	669

Notes: Intention to Treat estimates from OLS regressions. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses. Column (1) reports the recruiter's need to see the CV (on a 7-point Likert scale), while Column (2) reports the normalized time taken by the recruiter to read and evaluate the cover letter. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.2.3 Recruiters Heterogeneity

Table A23: Female Recruiters: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.13 (0.11)	-0.17 (0.12)	-0.13 (0.12)	-0.07 (0.11)	-0.13 (0.12)	-0.12 (0.11)	-0.20 (0.20)	-0.25 (0.20)
Full	0.05 (0.11)	0.07 (0.12)	0.13 (0.11)	0.07 (0.12)	0.11 (0.12)	0.09 (0.11)	0.12 (0.20)	0.19 (0.20)
Cover Letter Written with GPT	-0.05 (0.09)	0.04 (0.10)	0.02 (0.10)	-0.01 (0.09)	-0.02 (0.09)	-0.06 (0.09)	-0.14 (0.18)	-0.18 (0.19)
Partial $\times$ Written with GPT	0.12 (0.13)	0.21 (0.14)	0.10 (0.14)	0.03 (0.12)	0.13 (0.13)	0.09 (0.13)	0.24 (0.24)	0.31 (0.27)
Full $\times$ Written with GPT	0.05 (0.14)	0.07 (0.13)	0.04 (0.13)	0.02 (0.13)	-0.03 (0.13)	0.05 (0.13)	-0.11 (0.26)	-0.21 (0.27)
t-test Partial: ChatGPT vs. No	0.47	0.02	0.25	0.76	0.29	0.77	0.42	0.50
t-test Full: ChatGPT vs. No	0.96	0.30	0.57	0.90	0.59	0.93	0.26	0.06
t-test Partial vs. Full: No ChatGPT	0.03	0.03	0.00	0.04	0.02	0.01	0.25	0.21
t-test Partial vs. Full: Yes ChatGPT	0.22	0.38	0.05	0.17	0.42	0.07	0.86	0.68
Control Group Mean	0.04	0.01	0.04	0.01	0.01	0.01	3.43	0.54
Control Group S.D.	0.99	1.01	1.00	0.99	1.01	1.01	1.20	0.50
Observations	1175	1175	1175	1175	1175	1175	1175	1175

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of only Female recruiters. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A24: Male Recruiters: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	0.14 (0.14)	0.03 (0.13)	0.20 (0.12)	0.09 (0.14)	0.11 (0.15)	0.05 (0.14)	-0.03 (0.24)	-0.05 (0.24)
Full	0.08 (0.14)	0.09 (0.13)	0.11 (0.13)	0.13 (0.14)	0.06 (0.14)	0.11 (0.13)	0.28 (0.24)	0.35 (0.24)
Cover Letter Written with GPT	-0.03 (0.10)	0.03 (0.11)	-0.01 (0.08)	0.01 (0.09)	0.00 (0.10)	-0.07 (0.09)	-0.03 (0.20)	0.05 (0.22)
Partial $\times$ Written with GPT	-0.08 (0.15)	0.03 (0.15)	-0.09 (0.13)	-0.01 (0.14)	-0.06 (0.15)	0.01 (0.14)	-0.03 (0.29)	-0.12 (0.32)
Full $\times$ Written with GPT	-0.01 (0.14)	-0.03 (0.15)	-0.13 (0.13)	0.02 (0.13)	-0.11 (0.14)	-0.08 (0.13)	-0.22 (0.27)	-0.31 (0.30)
t-test Partial: ChatGPT vs. No	0.28	0.64	0.32	0.99	0.59	0.62	0.87	0.77
t-test Full: ChatGPT vs. No	0.66	0.92	0.19	0.76	0.28	0.14	0.30	0.26
t-test Partial vs. Full: No ChatGPT	0.77	0.47	0.21	0.36	0.39	0.98	0.16	0.07
t-test Partial vs. Full: Yes ChatGPT	0.92	0.81	0.28	0.41	0.39	0.62	0.56	0.38
Control Group Mean	-0.06	-0.02	-0.05	-0.02	-0.01	-0.02	3.36	0.52
Control Group S.D.	1.02	0.98	0.99	1.02	0.99	0.98	1.20	0.50
Observations	830	830	830	830	830	830	830	830

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of only Male recruiters. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A25: Old Recruiters: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.03 (0.13)	-0.12 (0.14)	-0.09 (0.14)	-0.06 (0.13)	-0.04 (0.14)	-0.09 (0.14)	-0.18 (0.21)	-0.29 (0.21)
Full	0.14 (0.15)	0.12 (0.15)	0.13 (0.15)	0.14 (0.15)	0.15 (0.16)	0.18 (0.14)	0.20 (0.23)	0.14 (0.22)
Cover Letter Written with GPT	-0.04 (0.10)	0.10 (0.12)	0.01 (0.11)	-0.01 (0.09)	0.01 (0.10)	-0.07 (0.10)	-0.15 (0.20)	-0.18 (0.21)
Partial $\times$ Written with GPT	0.05 (0.15)	0.05 (0.15)	-0.01 (0.15)	0.04 (0.13)	0.06 (0.14)	0.10 (0.14)	0.16 (0.26)	0.20 (0.30)
Full $\times$ Written with GPT	0.07 (0.16)	0.02 (0.16)	0.05 (0.16)	0.10 (0.15)	-0.00 (0.16)	0.03 (0.16)	0.10 (0.29)	0.18 (0.32)
t-test Partial: ChatGPT vs. No	0.99	0.20	0.98	0.84	0.55	0.84	0.92	0.90
t-test Full: ChatGPT vs. No	0.83	0.33	0.60	0.46	0.92	0.72	0.84	0.99
t-test Partial vs. Full: No ChatGPT	0.02	0.00	0.00	0.00	0.04	0.00	0.01	0.01
t-test Partial vs. Full: Yes ChatGPT	0.07	0.05	0.01	0.02	0.20	0.06	0.08	0.08
Control Group Mean	0.01	0.06	0.06	-0.00	0.03	0.02	3.40	0.54
Control Group S.D.	1.06	1.02	1.05	1.03	1.02	1.05	1.26	0.50
Observations	945	945	945	945	945	945	945	945

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of only recruiters above the median age. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Table A26: Young Recruiters: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.01 (0.11)	-0.04 (0.12)	0.09 (0.12)	0.04 (0.12)	-0.01 (0.12)	-0.01 (0.11)	-0.09 (0.22)	-0.06 (0.23)
Full	0.02 (0.11)	0.07 (0.11)	0.14 (0.11)	0.08 (0.11)	0.06 (0.11)	0.05 (0.11)	0.21 (0.21)	0.35* (0.21)
Cover Letter Written with GPT	-0.03 (0.08)	-0.03 (0.09)	0.00 (0.08)	0.02 (0.09)	-0.03 (0.09)	-0.05 (0.08)	-0.04 (0.18)	-0.01 (0.20)
Partial $\times$ Written with GPT	0.03 (0.13)	0.25* (0.13)	0.08 (0.14)	0.01 (0.13)	0.05 (0.13)	0.03 (0.13)	0.12 (0.26)	0.09 (0.30)
Full $\times$ Written with GPT	-0.03 (0.12)	0.03 (0.12)	-0.11 (0.11)	-0.05 (0.12)	-0.13 (0.12)	-0.05 (0.11)	-0.40* (0.24)	-0.59** (0.25)
t-test Partial: ChatGPT vs. No	0.97	0.04	0.41	0.80	0.87	0.82	0.49	0.72
t-test Full: ChatGPT vs. No	0.52	0.97	0.26	0.70	0.09	0.27	0.05	0.00
t-test Partial vs. Full: No ChatGPT	0.99	0.68	0.57	0.76	0.98	0.76	1.00	0.57
t-test Partial vs. Full: Yes ChatGPT	0.77	0.41	0.16	0.90	0.34	0.84	0.15	0.23
Control Group Mean	-0.01	-0.06	-0.06	0.00	-0.03	-0.02	3.40	0.52
Control Group S.D.	0.94	0.98	0.95	0.97	0.98	0.95	1.15	0.50
N	1060	1060	1060	1060	1060	1060	1060	1060

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of only recruiters below the median age. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.2.4 Partial Information Only - Perceptions of Recruiters

In this section, we replicate Table 3 among recruiters in the *Partial Information* treatment, with the independent variable being the recruiter’s belief of whether the cover letter was written with or without the assistance of LLMs. While the independent variable is not exogenous and hence causality cannot be claimed, the results provide suggestive evidence that recruiters evaluate cover letters that they perceive were written with the use of LLMs more positively, however this does not translate into a higher interview likelihood.

Table A27: Partial Info Only: Effect of Perception About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Thinks CL Written With GPT	0.15* (0.08)	0.28*** (0.08)	0.20** (0.08)	0.07 (0.08)	0.05 (0.09)	0.18** (0.08)	0.19 (0.16)	0.16 (0.18)
Control Group Mean	-0.07	-0.14	-0.08	-0.03	-0.04	-0.10	3.32	0.49
Control Group S.D.	0.99	0.95	0.99	0.92	0.99	0.97	1.16	0.50
Observations	660	660	660	660	660	660	660	660

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter’s age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (when recruiters thought the cover letter was not written with LLM assistance). Robust standard errors are in parentheses. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Panel A reports treatment effects for all cover letters, while Panels B and C report treatment effect for cover letters written without LLM, and with LLM assistance, respectively. Sample consists of only recruiters in the *Partial Information* treatment. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.2.5 Tercile Heterogeneity

Table A28 reports treatment effects for cover letters in the lower tercile. The evaluations of the recruiters in the recruiter-side experiment are similar to those of recruiters in the job-seeker Experiment, as the average evaluation of these cover letters is below the mean, indicated by the negative value of the standardized control group mean (referring to the *Partial Info* treatment). Cover letters written with the assistance of ChatGPT are scored higher than those without (comparing the Control Group Means in Panels B vs. C), in line with the findings from Figure 1 which illustrates that the positive treatment effects of ChatGPT assistance on a cover letter’s quality are primarily driven by lower-quality applicants. Furthermore, complete information on whether the applicant used ChatGPT or not does not impact either the evaluation of the quality of the cover letter (Columns 1 - 6), nor the likelihood of inviting the candidate to a job interview (Columns 7-8). Therefore, revealing the LLM assistance status of the applicant does not have an effect on the recruiters’ evaluations of the low-quality cover letters.

Table A28: Main Results: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation - Lower Tercile

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	0.13 (0.15)	0.07 (0.15)	0.03 (0.14)	0.09 (0.15)	0.12 (0.15)	0.04 (0.15)	0.28 (0.27)	0.19 (0.29)
Full	0.05 (0.14)	0.06 (0.14)	0.07 (0.13)	0.07 (0.13)	0.03 (0.14)	0.07 (0.14)	0.24 (0.25)	0.20 (0.29)
Cover Letter Written with GPT	0.44*** (0.12)	0.40*** (0.13)	0.34*** (0.12)	0.38*** (0.11)	0.38*** (0.12)	0.48*** (0.13)	0.86*** (0.23)	0.83*** (0.25)
Partial × Written with GPT	-0.18 (0.18)	-0.09 (0.18)	-0.06 (0.18)	-0.09 (0.17)	-0.23 (0.18)	-0.14 (0.18)	-0.42 (0.33)	-0.38 (0.38)
Full × Written with GPT	-0.01 (0.17)	-0.00 (0.17)	0.02 (0.16)	0.03 (0.16)	-0.10 (0.17)	-0.04 (0.17)	-0.28 (0.31)	-0.29 (0.36)
t-test Partial: ChatGPT vs. No	0.04	0.02	0.03	0.02	0.22	0.01	0.06	0.11
t-test Full: ChatGPT vs. No	0.00	0.00	0.00	0.00	0.04	0.00	0.01	0.05
t-test Partial vs. Full: No ChatGPT	1.00	0.67	0.36	0.72	0.74	0.47	0.90	0.76
t-test Partial vs. Full: Yes ChatGPT	0.49	0.51	0.29	0.48	0.84	0.35	0.62	0.69
Control Group Mean	-0.40	-0.39	-0.30	-0.36	-0.35	-0.34	2.92	0.38
Control Group S.D.	1.05	1.04	1.01	1.00	1.04	1.05	1.25	0.49
N	669	669	669	669	669	669	669	669

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter’s age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of cover letters that were identified as being part of the lower tercile, based on evaluations from the first Experiment. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

Focusing next on the medium-quality cover letters - those which were graded in the middle

tercile of the job-seeker experiment - Table A29 highlights heterogeneity in the evaluations of cover letters written with and without the assistance of LLMs. The perceived quality of the cover letter is unchanged as a result of *Full Information*, as the average grade of the cover letter is the same for both types of cover letters (see the Control Group Mean in Panels B and C), and the treatment effects are indistinguishable in Columns 1-6 across Panels A-C. Despite evaluating the cover letters as equally good, recruiters were statistically significantly more likely to recommend the applicant to the next step of the application process. This can be seen by comparing Columns 7 and 8 across Panels B and C. Informing recruiters that a cover letter was written without the assistance of ChatGPT statistically significantly increased the likelihood and chance of inviting the applicant to a job interview. Similarly, informing the recruiter that a cover letter was written with the assistance of ChatGPT reduced the likelihood of inviting the applicant to a job interview, albeit not statistically significantly. This suggests that recruiters place a premium on applications that did not use ChatGPT, even if the cover letters are deemed to be of comparable, average quality.

Table A29: Main Results: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation - Medium Tercile

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Cover Letter Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.07 (0.13)	-0.12 (0.13)	-0.01 (0.14)	-0.07 (0.13)	-0.05 (0.13)	0.01 (0.12)	-0.31 (0.23)	-0.19 (0.27)
Full	0.03 (0.13)	0.05 (0.13)	0.10 (0.14)	0.06 (0.13)	0.16 (0.13)	0.15 (0.12)	0.18 (0.24)	0.40 (0.28)
Cover Letter Written with GPT	-0.09 (0.11)	-0.02 (0.11)	-0.10 (0.11)	-0.10 (0.11)	0.06 (0.11)	0.01 (0.11)	-0.22 (0.23)	-0.21 (0.26)
Partial $\times$ Written with GPT	0.11 (0.16)	0.24 (0.16)	0.10 (0.17)	0.12 (0.16)	0.08 (0.16)	-0.02 (0.16)	0.39 (0.33)	0.33 (0.38)
Full $\times$ Written with GPT	0.05 (0.16)	0.03 (0.16)	-0.02 (0.17)	0.07 (0.16)	-0.22 (0.15)	-0.09 (0.15)	-0.17 (0.34)	-0.47 (0.39)
t-test Partial: ChatGPT vs. No	0.90	0.07	1.00	0.88	0.24	0.92	0.57	0.66
t-test Full: ChatGPT vs. No	0.75	0.90	0.37	0.80	0.17	0.50	0.11	0.01
t-test Partial vs. Full: No ChatGPT	0.42	0.48	0.57	0.21	0.42	0.16	0.20	0.37
t-test Partial vs. Full: Yes ChatGPT	0.73	0.73	0.97	0.51	0.50	0.48	0.87	0.45
Control Group Mean	0.22	0.21	0.17	0.16	0.23	0.23	3.62	0.59
Control Group S.D.	0.93	0.91	0.93	0.97	0.92	0.92	1.14	0.49
N	668	668	668	668	668	668	668	668

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). Sample consists of cover letters that were identified as being part of the medium tercile, based on evaluations from the first Experiment. ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

A.2.6 No Clustered Standard Errors

Table A30: Main Results: Effect of Knowledge About ChatGPT Usage on Cover Letter Evaluation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total	Layout	Intro	Experience	Motivation	Closing	Likelihood of Interview	High Chance of Interview
Partial	-0.03 (0.08)	-0.10 (0.08)	-0.01 (0.08)	-0.02 (0.08)	-0.04 (0.08)	-0.05 (0.08)	-0.14 (0.15)	-0.16 (0.16)
Full	0.06 (0.08)	0.08 (0.08)	0.12 (0.08)	0.10 (0.08)	0.09 (0.08)	0.10 (0.08)	0.19 (0.15)	0.26* (0.16)
Cover Letter Written with GPT	-0.04 (0.08)	0.03 (0.08)	0.01 (0.08)	0.00 (0.08)	-0.01 (0.08)	-0.06 (0.08)	-0.10 (0.15)	-0.09 (0.16)
Partial $\times$ Written with GPT	0.04 (0.11)	0.14 (0.11)	0.03 (0.11)	0.02 (0.11)	0.05 (0.11)	0.06 (0.11)	0.14 (0.20)	0.14 (0.22)
Full $\times$ Written with GPT	0.02 (0.11)	0.02 (0.11)	-0.04 (0.11)	0.02 (0.11)	-0.07 (0.11)	-0.01 (0.11)	-0.15 (0.20)	-0.25 (0.22)
t-test Partial: ChatGPT vs. No	0.99	0.02	0.64	0.78	0.57	0.99	0.61	0.76
t-test Full: ChatGPT vs. No	0.82	0.45	0.71	0.77	0.29	0.34	0.12	0.03
t-test Partial vs. Full: No ChatGPT	0.12	0.03	0.08	0.03	0.19	0.03	0.08	0.03
t-test Partial vs. Full: Yes ChatGPT	0.31	0.41	0.40	0.11	0.92	0.28	0.82	0.80
Control Group Mean	0.00	0.00	0.00	0.00	0.00	0.00	3.40	0.53
Control Group S.D.	1.00	1.00	1.00	1.00	1.00	1.00	1.20	0.50
N	2005	2005	2005	2005	2005	2005	2005	2005

Notes: Intention to Treat estimates. Column 1-6 report treatment effects from OLS regressions, while columns 7 and 8 report multinomial logit and logit regressions, respectively. All regressions include controls for the recruiter's age and sex, as well as job-applicant fixed effects. Standard errors are clustered at the recruiter level. Control mean refers to the mean value of the outcome in the control group (No Information). Robust standard errors are in parentheses, and are clustered at the recruiter-level. Column 1-7 refer to variables as described in Appendix B.1.3, while Column 8 refers to a dummy if the Likelihood of Being Invited to an Interview is greater than three (on a five-point scale). ***, ** and * represent significant differences at the 1, 5 and 10% level, respectively.

B Experiment Logistics

B.1 Job-seeker Experiment

B.1.1 Blocked Websites

OpenAI	Perplexity	Gemini
Falcon LLM	Google Gemini	Huggingface
Mistral	Llama	Claude
Anthropic	CopyAI	Anyword
Sudowrite	Writer	Writesonic
Rytr	Jasper AI	Simplified
Wordai	Grammarly	Careerflow AI
Resume IO	Kickresume	Teal HQ
Google Drive	Google Mail	Dropbox
Onedrive	SurfDrive	Rezi AI Cover Letter Builder
Jobscan	Coverletter Copilot	Microsoft Copilot
Myperfectresume	Coverdoc AI	AI Apply
Lazyapply	Zety	Aicoverlettergenerator
Coverletter-ai	Easycoverletter	Master Interview AI
There's an AI for that		

Table A31: List of Blocked Domain Names

OpenAI, in bold, is the only domain page that is blocked in the Control group, but not in the Treatment group.

B.1.2 Training Materials

The training material for the Control group can be found [here](#).

The training material for the Treatment group can be found [here](#).

B.1.3 Evaluation Criteria

	Low (0-4)	Medium (4-7)	High (7-10)
Layout, Writing Quality, and Clarity	Not professional in appearance; poor formatting; No flow or order to the way things are discussed; spelling and grammar errors; confusing sentences or main points	Letter generally looks clear and professional; Generally able to follow organization and flow; very few mistakes in spelling and grammar; some connection between the narrative and the position, but not maximized	Professional appearance; clean fonts and formatting; Clear organization; clean and consistent layout; free of grammar, spelling errors; overall narrative that clearly connects main points
Introduction	Does not cover basic info; weak link to the body of the cover letter; no link between the applicant and the position.	Conveys basic information in an unengaging form. Links to the main part of the cover letter, but not very effectively.	Covers basic information, offers an engaging and gripping way into the body of the letter and clearly connects the person to the position.
Relevant Experiences and Demonstration of Skills	not enough/ too much information in key areas; background, education and experience not fully explained; more questions raised than answered; all assertions without foundation or specifics to support them.	background, education and experience laid out but not connected to the position; Vague but inconsistent narrative.	Clear narrative; outlines background, education and experience fully and with specifics that connect directly to the position.
Motivation for Role and Alignment with Job	Little discussion of their motivation, the company, or how they would fit into existing organization.	Decent, but not always coherent motivation for the role; some good connections between the position and the applicant; some research done about the firm, but not that much.	focus on how they fit the job and will be effective members of the organization; outlines motivation in a coherent manner; strong understanding of the role, the organization, and how they would fit within the organization.
Closing	Not a strong closing statement; repetitive or wandering; no clear 'final message' to the reader and how they fit the job as outlined.	Has a sense of a closing statement but unenthusiastic or unconvincing; tries to convey too much, making it confusing; no succinct final message.	Strong closing statement of purpose; clearly outlines how their background, education and experience have prepared them for this specific position (without being repetitive).

Figure A8. Cover Letter Evaluation Criteria

	Low (0-4)	Medium (4-7)	High (7-10)
Layout, Writing Quality, and Clarity	Not professional in appearance; poor formatting; No flow or order to the way things are discussed; spelling and grammar errors; confusing sentences or main points.	CV generally looks clear and professional; Generally able to follow organization and flow; very few mistakes in spelling and grammar; descriptions are generally clear.	Professional appearance; clean fonts and formatting; Clear organization; clean and consistent layout; free of grammar, spelling errors; effective use of CV phrasing.
Education	Below-average student, or from a subject not related to position they are applying for.	Average student, however does not belong to the top share of their cohort.	One of the top students in their cohort, as part of a demanding and relevant degree.
Experience	No or little relevant work / internship, or leadership experience.	Some relevant work / internship, or leadership experience, however it could be more.	Strong relevant work / internship, or leadership experience, for example through work experience or student associations.
Extra-curricular	No evidence of involvement in extra-curricular activities.	Some involvement in extra-curricular activities.	Strong involvement in extra-curricular activities.

Figure A9. CV Evaluation Criteria

B.2 Recruiter Experiment

Figure A10. Cover Page

(a) No Info	(b) Partial Info	(c) Full Info
 In the spring of 2024, university students wrote a cover letter for a Traineeship at Rabobank, one of the largest banks in the Netherlands. You will be asked to evaluate FIVE cover letters, based on a pre-defined grading rubric. Please grade them as if you were a recruiter of the company the students are applying to. The job description is the following:	 In the spring of 2024, university students wrote a cover letter for a Traineeship at Rabobank, one of the largest banks in the Netherlands. You will be asked to evaluate FIVE cover letters, based on a pre-defined grading rubric. Please grade them as if you were a recruiter of the company the students are applying to. Some of the Cover Letters were written with the assistance of ChatGPT, while others were NOT. You will be asked to indicate whether you think the Cover Letter was written using the help of ChatGPT, or not. The job description is the following:	 In the spring of 2024, university students wrote a cover letter for a Traineeship at Rabobank, one of the largest banks in the Netherlands. You will be asked to evaluate FIVE cover letters, based on a pre-defined grading rubric. Please grade them as if you were a recruiter of the company the students are applying to. Some of the Cover Letters were written with the assistance of ChatGPT, while others were NOT. We WILL indicate whether the Cover Letter was written using the help of ChatGPT, or not, before you read the cover letter. The job description is the following:

Figure A11. Cover Letter Page

(a) No Info	(b) Partial Info	(c) Full Info
 Below you will find a cover letter, written for the job description you just saw, which you can find here. You will have to evaluate the cover letter, based on the evaluation rubric, which you can find here.	 Below you will find a cover letter, written for the job description you just saw, which you can find here. You will have to evaluate the cover letter, based on the evaluation rubric, which you can find here.	 Below you will find a cover letter, written for the job description you just saw, which you can find here. While writing the cover letter, the student WAS NOT allowed to use ChatGPT. You will have to evaluate the cover letter, based on the evaluation rubric, which you can find here.

C Recruiter Insights on the Labor Market

The recruiter-side survey was also used to gain further insights into recruiters’ perceptions of LLMs, and cover letters. These are outlined below:

Recruiters were asked to evaluate the sub-components of the cover letter (Layout, Introduction, Experience, Motivation, Conclusion) on a scale of 1-5, where 1 indicated the most personalized part of the cover letter, while 5 indicated the least personalized part of the cover letter.

Table A32: Personalization of Cover Letter Components

Cover Letter Component	Degree of Personalization
Layout	3.72
Introduction	2.46
Experience	2.44
Motivation	2.24
Conclusion	4.15

This indicates that the most personalized sections of the cover letter are the Motivation and Introduction. The least personalized section is the Conclusion.

Recruiters were also asked, on a five-point Likert scale, the degree to which they agree or disagree with the following statements: “It is acceptable for job applicants to use LLMs in their application”; “Job applicants should disclose LLM usage in their applications”; “The use of LLMs (e.g., ChatGPT) in a cover letter improves their grammar”; “The use of LLMs (e.g., ChatGPT) in a cover letter increases its originality and personalization”. The results are presented in the table below:

Table A33: Agree-ability with Statements

	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree	Average Value
Acceptable to Use	7.48%	24.19%	18.95%	40.40%	8.98%	3.19
Should Disclose Usage	8.23%	16.46%	26.1%8	27.68%	21.45%	3.38
LLM improves Grammar	2.74%	11.22%	17.21%	48.13%	20.70%	3.73
LLM increases originality	24.19%	38.40%	20.45%	13.72%	3.24%	2.33

Recruiters were asked on a five-point Likert scale how confident they were that they could detect LLM usage by job applicants. The results are presented in the Table below:

Table A34: Recruiter Confidence Detecting Use of LLMs in Cover Letters

	Not	Slightly	Moderately Confident	Very	Extremely	Average Value
Confident Detect LLM Use	5.51%	34.09%	42.61%	15.54%	2.26%	2.75

Recruiters were also asked about their attitudes towards applicants using LLMs in their cover letters:

Table A35: Recruiter perceptions about Use of LLMs in Cover Letters

	Very Negative	Negative	Neutral	Positive	Very Positive	Average Value
LLM Use in Cover Letter	7.29%	23.87%	44.72%	21.86%	2.26%	2.88

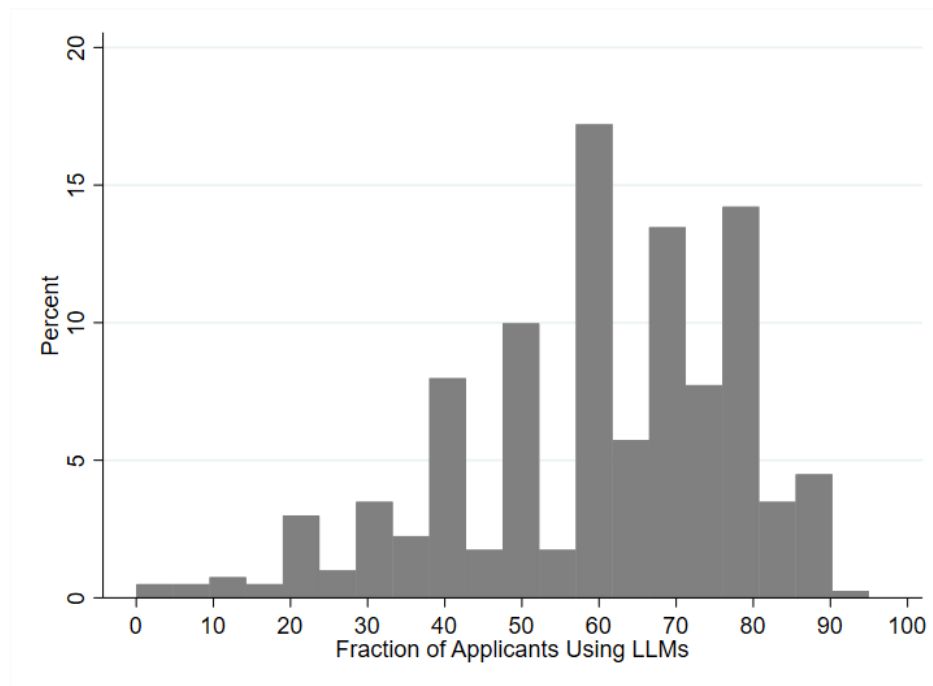
Recruiters were asked whether they thought LLMs like ChatGPT would have a smaller or greater impact on the quality of job applications, compared with Algorithmic Writing Assistants like Grammarly, which were empirically evaluated by [Wiles et al. \(2025\)](#).

Table A36: Impact of LLMs vs. Algorithmic Writing Assistants

	Far Smaller	Somewhat Smaller	Equally- sized	Somewhat Larger	Far Larger	Average Value
Impact of LLMs vs. AWA	3.99%	7.98%	19.20%	48.13%	20.70%	3.74

Lastly, recruiters were asked what percentage of applicants they think currently use LLMs in their job applications. The mean was 60.63%, with a wide distribution as illustrated by Figure A12.

Figure A12. Histogram of Recruiter's Perception on LLM Use Among Applicants



D ChatGPT Conversation Analysis

This section details our methodology for analyzing conversations between users and ChatGPT during cover letter writing. We use OpenAI API with model *gpt-4o-mini* to classify and quantify user interactions, enabling nuanced analysis beyond simple keyword matching.

D.1 Methodology

We analyze six key cover letter components, classifying user messages into three categories:

- **None:** Message does not pertain to the specific section.
- **Content:** User seeks substantive content related to the section.
- **Guidance:** User requests structural/formatting advice.

Our analysis covers these sections:

Layout: Formatting, structure, presentation, grammar, readability

Introduction: Opening statements, salutations, position identification

Experience: Professional background, education, skills, qualifications

Motivation: Personal drive, interest in position/company, career goals

Conclusion: Closing statements, next steps, gratitude, contact information

Generic: Broad requests like "write me a cover letter" without section-specific focus

Additionally, we analyze four meta-categories:

Company Research: Questions about company info, values, culture, industry

General Strategy: Overall approach, structure planning, best practices

Formatting: Length, font, spacing, visual presentation

Revision Requests: Improvements, editing, refining, changes

D.2 API Implementation

We developed a Python script that interfaces with the OpenAI API. The system prompt instructs:

```
1 {  
2     "role": "system",  
3     "content": "Analyze this conversation between a user and ChatGPT about cover  
    ↪ letter writing.
```

```

4
5 For each section below, classify the user's engagement:
6
7 SECTIONS:
8 - layout: Formatting, structure, presentation, grammar, readability
9 - introduction: Opening statements, salutations, position identification
10 - experience: Professional background, education, skills, qualifications
11 - motivation: Personal drive, interest in position/company, career goals
12 - conclusion: Closing statements, next steps, gratitude, contact info
13 - generic: Broad requests like \"write me a cover letter\", general help without
    ↪ specific section focus
14
15 For each section, classify as:
16 - \"none\": No engagement with this section
17 - \"guidance\": User asks for advice/tips on how to approach the section
18 - \"content\": User asks ChatGPT to generate or improve actual text for the
    ↪ section
19
20 Return ONLY a JSON object:
21 {
22   \"layout\": \"none|guidance|content\",
23   \"introduction\": \"none|guidance|content\",
24   \"experience\": \"none|guidance|content\",
25   \"motivation\": \"none|guidance|content\",
26   \"conclusion\": \"none|guidance|content\",
27   \"generic\": \"none|guidance|content\"
28 }
29
30 Base analysis only on USER messages.\"
31 }

```

```

1 \"role\": \"system\",
2   \"content\": \"Analyze if this user engaged with these meta-categories during
    ↪ cover letter writing:

```

```
3
4 CATEGORIES:
5 - company_research: Questions about company info, values, culture, industry
6 - general_strategy: Overall approach, structure planning, best practices
7 - formatting: Length, font, spacing, visual presentation
8 - revision_requests: Improvements, editing, refining, changes
9
10 Return ONLY a JSON object:
11 {
12   \"company_research\": true|false,
13   \"general_strategy\": true|false,
14   \"formatting\": true|false,
15   \"revision_requests\": true|false"
16 }
```

D.3 Additional Analysis

D.3.1 Meta-Category Engagement

Figure A13. Meta-Category Engagement

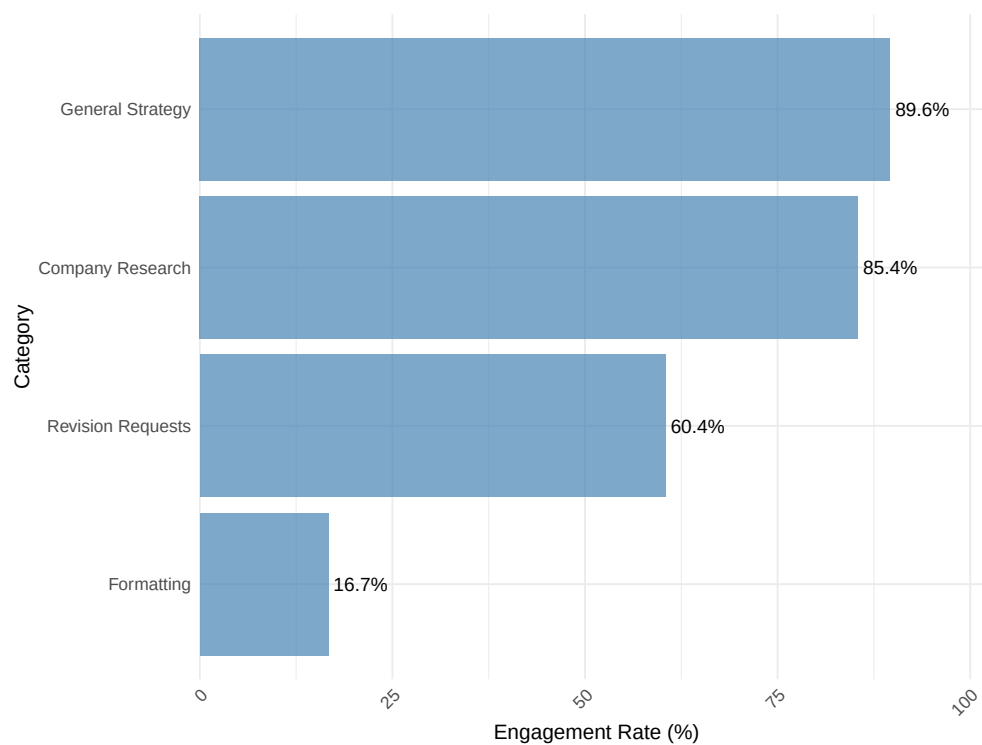


Figure A13 displays the percentage of students who engaged with each meta-category during their ChatGPT interactions for cover letter writing.

D.3.2 Message Distribution

Table A37: Summary Statistics for Number of User Messages

Agent	Mean	Median	Variance	SD	Min	Max
User	6.416667	5	19.22695	4.384855	1	16

Figure A14. Distribution of User Messages per Conversation

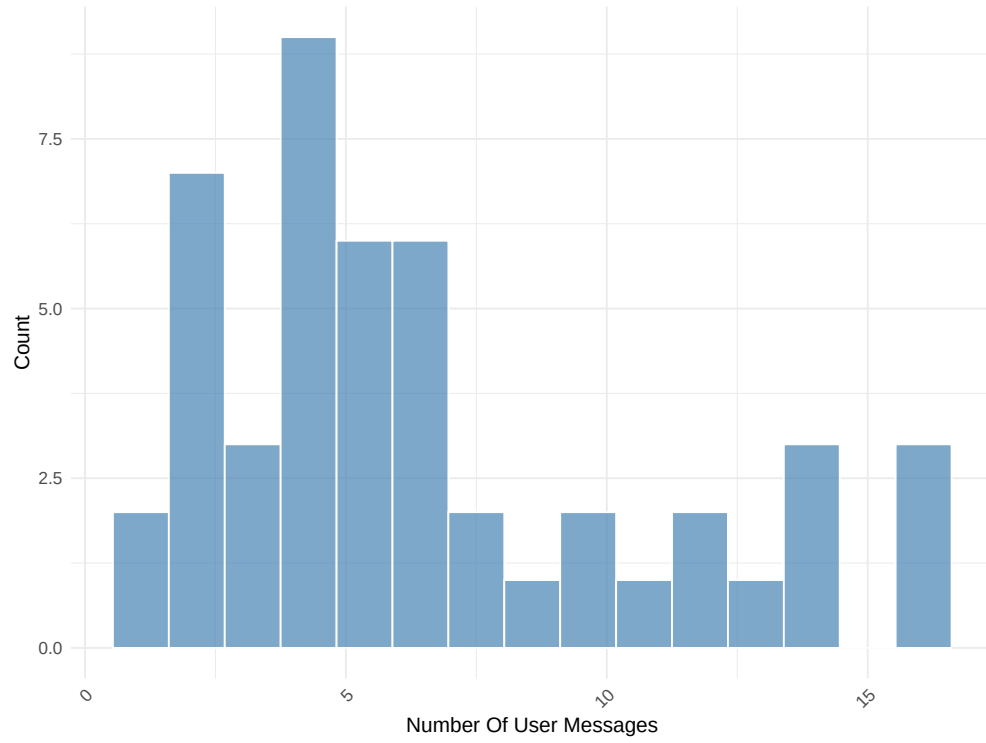


Figure A14 shows the frequency distribution of messages exchanged between users and ChatGPT, demonstrating substantial variation in user engagement patterns.

E Model Details, Proofs and Extension

This section provides further details on the model. In addition to providing a proof of proposition 1 in the main text; we provide proofs of some properties of assignment models in the case of a continuum of agents. Many of these properties—such as uniqueness of assortative matching and optimality of positive assortative matching with supermodularity—are well-known in the case of discrete number of agents. We provide details on the extensions of these results to the case of continuum of agents; to our knowledge we are the first to do so systematically.

E.1 Perfect Information Economy

The static economy consists of a continuum of workers, indexed by their quality s and distributed with CDF G_s , and a continuum of firms, indexed by their quality x and distributed with CDF G_x . We assume that both G_s and G_x admit densities denoted by g_x and g_s and have positive and bounded supports $\mathcal{S} := [\underline{s}, \bar{s}]$ and $\mathcal{X} := [\underline{x}, \bar{x}]$. The value of a match between a worker of type s and a firm of type x is given by $f(x, s)$ which satisfies Assumption 1:

Assignment in the economy without information frictions is defined by a function $x = \sigma(s)$, which matches each worker s with a firm x . In equilibrium, all firms and workers must be matched with each other, and matches are one-to-one: as explained above, there are no multi-worker firms or multi-firm workers. Therefore, the key property of σ is that, when considered as a set transformation, it must match an equal mass of workers and firms, in other words, σ must be a *measure-preserving* transformation. Given an assignment function σ , the aggregate value (or welfare) in this economy with perfect information is given by

$$V_P = \int_{\mathcal{S}} f(\sigma(s), s) dG_s(s) \quad (5)$$

The monotonically-increasing assignment function that assigns top firms with top workers and bottom firms with bottom workers is known as *positively assortative matching* (PAM). PAM essentially matches the top $q = G_x(x)$ quantile of firms with the top $q = G_s(s)$ quantile of workers, and thus is given by $x = \sigma_P(s) = G_x^{-1}(G_s(s))$. Lemma 1 shows that PAM is unique in that it is the only monotonically-increasing measure-preserving transformation from the set of workers to the set of firms.

Lemma 1. *PAM is the only monotonically-increasing assignment function.*

Proof of Lemma 1. The relevant probability spaces are $(\mathcal{S}, \mathcal{B}_s, P_s)$ and $(\mathcal{X}, \mathcal{B}_x, P_x)$ where $\mathcal{B}_s, \mathcal{B}_x$ are

the Borel σ -algebras of $\mathcal{S}, \mathcal{X}$, and P_s, P_x are the probability measures associated with distribution functions G_s, G_x .

An assignment function, as explained in the main text, is a function $\sigma : \mathcal{S} \rightarrow \mathcal{X}$ that assigns each worker $s \in \mathcal{S}$ to a firm $x \in \mathcal{X}$. Considered as a set transformation, it is a mapping from $\mathcal{B}_s$ to $\mathcal{B}_x$ that assigns groups of workers $S \subset \mathcal{S}$ to groups of firms $X \subset \mathcal{X}$. In equilibrium, an assignment transformation must be measure-preserving, i.e.

$$P_s\{\sigma^{-1}(B)\} = P_x\{B\} \quad \forall B \in \mathcal{B}_x \quad (6)$$

Consider the sets $B = (\underline{x}, t)$ of the Borel σ -algebra $\mathcal{B}_x$, for any $\underline{x} < t \leq \bar{x}$. If σ is monotonically increasing, i.e. $\sigma(t') > \sigma(t) \quad \forall t' > t$, this implies that:

$$P_s\{\sigma^{-1}((\underline{x}, t))\} = P_s\{(\underline{s}, \sigma^{-1}(t))\} = G_s(\sigma^{-1}(t)). \quad (7)$$

Where we used the fact that $\sigma^{-1}(\underline{x}) = \underline{s}$, since otherwise $\underline{x}$ or $\underline{s}$ will be unmatched, violating preservation of measure. Then Equation (6) gives:

$$G_s(\sigma^{-1}(t)) = G_x(t)$$

implying that σ must be PAM: $\sigma(t) = G_x^{-1}(G_s(t))$.

When dealing with monotonically-decreasing assignments, Equation (7) becomes

$$P_s\{\sigma^{-1}((\underline{x}, t))\} = P_s\{(\sigma^{-1}(t), \bar{s})\} = 1 - G_s(\sigma^{-1}(t)) \quad (8)$$

which leads to *negative assortative matching*: $G_x^{-1}(1 - G_s(t))$ being the unique monotonically-decreasing assignment. ■

In the context of discrete distributions [Becker \(1973\)](#) has shown that under the assumption of complementarity, PAM is the optimal assignment that maximizes Equation (5). We briefly extend this to the case of a continuum of workers and firms.

Lemma 2. *The positive assortative matching $\sigma_P(s) = G_x^{-1}(G_s(s))$ is optimal.*

Proof of Lemma 2. The relevant probability space is $(\mathcal{S}, \mathcal{B}_s, P_s)$ where $\mathcal{B}_s$ is the Borel σ -algebra of $\mathcal{S}$, and P_s is the probability measure associated with distribution function G_s .

For this proof we use the rearrangement inequality theorem of [Burchard and Hajaiej \(2006\)](#), who extend the results of [Crowe et al. \(1986\)](#), [Almgren and Lieb \(1989\)](#), and [Brock \(2000\)](#). This

theorem, a generalization of the Hardy-Littlewood inequality, states that if f is a supermodular function, the following inequality holds for any measurable functions u, v :

$$\int_{\mathcal{S}} f(u(s), v(s)) dP_s \leq \int_0^1 f(\bar{u}(z), \bar{v}(z)) dz \quad (9)$$

where $\bar{u}(s)$ is the unique non-increasing rearrangement of $u(s)$ defined as:

$$\bar{u}(z) := \sup \{t \geq 0 : \rho_u(t) \geq z\}$$

where $\rho_u(t) = P_s(\{s \in \mathcal{S} : u(s) > t\})$ is the distribution function of u (a similar definition holds for $\bar{v}(s)$).

Intuitively, a non-increasing rearrangement of a function u is function $\bar{u}$ whose level sets have the same measure as u but are re-arranged in a decreasing order. For more information on rearrangements and related inequalities see [Burchard \(2009\)](#).

In our case, $u(s) = G_x^{-1}(G_s(s))$ and $v(s) = s$. Thus

$$\rho_u(t) = P_s(\{s \in \mathcal{S} : G_x^{-1}(G_s(s)) > t\}) = P_s(\{s \in \mathcal{S} : s > G_s^{-1}(G_x(t))\}) \quad (10)$$

$$= 1 - G_x(t) \quad (11)$$

$$\rho_v(t) = P_s(\{s \in \mathcal{S} : s > t\}) = 1 - G_s(t) \quad (12)$$

So

$$\bar{u}(z) = \sup \{t \geq 0 : 1 - G_x(t) \geq z\} = G_x^{-1}(1 - z) \quad (13)$$

$$\bar{v}(z) = \sup \{t \geq 0 : 1 - G_s(t) \geq z\} = G_s^{-1}(1 - z) \quad (14)$$

so the rearrangement inequality states

$$\int_{\mathcal{S}} f(\sigma_P(s), s) dG_s(s) \leq \int_0^1 f(G_x^{-1}(1 - z), G_s^{-1}(1 - z)) dz \quad (15)$$

Let I denote the rightmost integral. With a simple change of variable $z = 1 - G_s(s)$ we obtain

$$I = - \int_{\bar{s}}^{\underline{s}} f(G_x^{-1}(G_s(s)), s) dG_s(s) = \int_{\underline{s}}^{\bar{s}} f(G_x^{-1}(G_s(s)), s) dG_s(s) \quad (16)$$

which shows that u, v are their own nonincreasing rearrangements, thus proving the optimality of PAM. ■

E.2 Imperfect Information without LLMs

Consider now an economy where firms cannot directly observe the quality of each worker and instead observe a signal y_1

$$y_1 = s + e \quad (17)$$

where e is the signal error with mean 0, distributed with CDF G_e on support $\mathcal{E} = [\underline{e}, \bar{e}]$. Here e reflects the fact that job-seekers' signal cover letter) does not fully reflect their relevant skills and abilities. Having received a signal y_1 , firms form their Bayesian estimates of the hidden worker type as $\hat{s}(y_1) = \mathbb{E}[s|y_1]$ (which has the same support $\mathcal{S}$ as the true distribution of workers).

Implicit in the signal Equation (17) is the assumption that signal values are independent of firm qualities and only depend on the worker quality. That is, we assume each worker only produces one signal, which is then observed by all firms. More intuitively this is the scenario where each worker producer one job application package and sends it to all firms (costs of job applications and posting vacancies are irrelevant to our framework).

After all firms have observed each worker's signal, they form a positive assortative matching based on estimated worker qualities $\hat{s}$. Thus the assignment of firms to estimated worker qualities is deterministic and given by $x = \sigma_I(\hat{s}) = G_x^{-1}(G_{\hat{s}}(\hat{s}))$, while the randomness in matching between firm and true worker qualities stems purely from the randomness in worker quality estimates due to the imperfect signal. Given a worker of type s , they will be matched according to their estimated quality $\hat{s}$ and so the expected output from the matches with this worker will be:

$$\mathbb{E}[f(\sigma_I(\hat{s}), s)|s] = \int_{\mathcal{S}} f(G_x^{-1}(G_{\hat{s}}(\hat{s})), s) dG_{\hat{s}|s}(\hat{s}|s)$$

This implies that the total output in this economy with PAM and imperfect information is given by:

$$V_I = \int_{\mathcal{S}} \int_{\mathcal{S}} f(G_x^{-1}(G_{\hat{s}}(\hat{s})), s) dG_{\hat{s}|s}(\hat{s}|s) dG_s(s) \quad (18)$$

The assignment given by σ_I is positively assortative with respect to the distribution of workers' quality estimates $\hat{s}$ and by Proposition 2 it is the optimal assignment between workers and firms given the information constraints. However, as asserted by Lemma 3, aggregate value in this economy with imperfect information is at most equal to the economy with perfect information.

Lemma 3. $V_I \leq V_P$

Proof of Lemma 3. With the concavity of f , Jensen's inequality shows that the value of the economy under imperfect information (V_I) is at most equal to that of the perfect information economy

(V_P):

$$V_I \leq \int_{\mathcal{S}} f(\tilde{\sigma}(s), s) dG_s(s) \leq \int_{\mathcal{S}} f(\sigma_P(s), s) dG_s(s) = V_P \quad (19)$$

where $\tilde{\sigma}(s) = \mathbb{E}[G_x^{-1}(G_{\hat{s}}(\hat{s}))|s] \neq \sigma_P(s)$ for all s a.s. and the second inequality follows from Proposition 2. ■

Due to the randomness in matching between firms and true worker qualities s , some firms and workers will be better off in the scenario with imperfect information compared to the perfect information case. However there will be aggregate losses due to concavity and complementarity of f : such a random matching will assign some lower-quality firms with some higher-quality workers and vice versa, thus deviating from the first-best assignment of σ_P . Because of the complementarity, these two cannot cancel out in the aggregate, resulting in net losses.

A completely random assignment will be the case in which there is no information of the form (17), i.e. workers do not produce any CVs or cover letters to provide information about their skills to the firms. Total value of this *random* assignment economy is given by

$$V_R = \int_{\mathcal{S}} \int_{\mathcal{X}} f(x, s) dG_x(x) dG_s(s) \quad (20)$$

where workers and firms are assigned to each other based on their (independent) distributions. Lemma 4 shows that total value in such an economy is at most equal to the imperfect information economy with informative signals.

Lemma 4. $V_R \leq V_I$

Proof of Lemma 4. Recall V_R is given by

$$V_R = \int_{\mathcal{S}} \int_{\mathcal{X}} f(x, s) dG_x(x) dG_s(s)$$

Using Jensen's inequality we get

$$V_R \leq \int_{\mathcal{S}} f(x_m, s) dG_s(s) =: I$$

where $x_m := \mathbb{E}[x]$ is the mean firm quality. The integral I can be simply re-written in a comparable form to V_I .

$$I = \int_{\mathcal{S}} \int_{\mathcal{S}} f(x_m, s) dG_{\hat{s}|s}(\hat{s}|s) dG_s(s) \quad (21)$$

In equation (21), we have the total value of an economy with the same information structure as

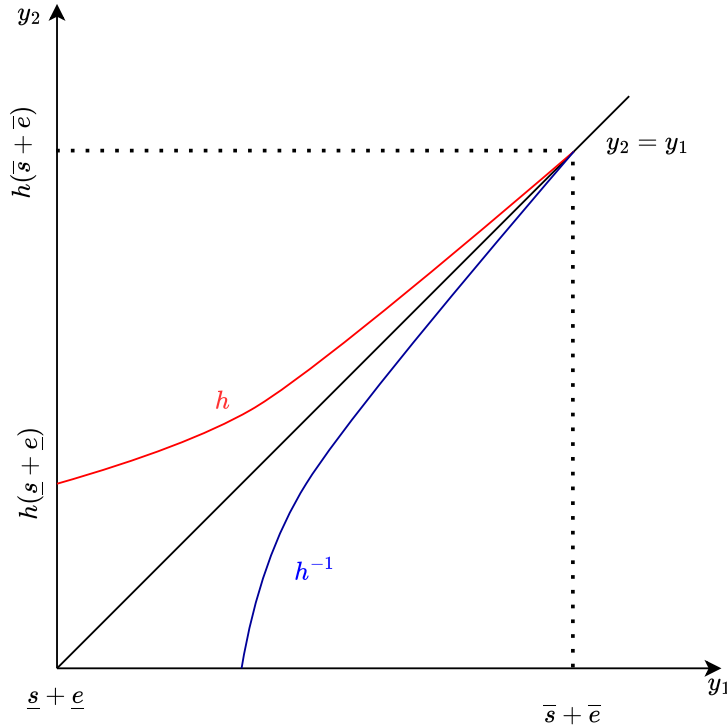
that of V_I , but all workers are matched with the mean firm. Since σ_I is the positive assortative matching and by Proposition 2 is the optimal assignment given the joint distribution of $\hat{s}$ and s , then we must have $V_R \leq V_I$. ■

E.3 Imperfect Information with LLMs

In this section we provide the proof of Proposition 1 in the main text, showing that total value in the economy with imperfect information and LLM-induced signal distortion-as described in the main text-is weakly lower than an analogous economy with imperfect information but without LLMs.

The effect of LLMs on the signals produced by workers in this economy is described by function h given in Assumption 2. The figure below shows an illustration of what such a function looks like:

Figure A15. Illustration function h



Notes: The function $h(y)$ captures the non-linear effect of LLMs on the signal y , with larger improvements for low-quality signals and smaller gains for high-quality ones.

Proof of Proposition 1. Since $\tilde{s}$ is a function of y , total value can be rewritten as an integral with

respect to the conditional distribution of signal values y

$$V_L = \int_{\mathcal{S}} \int_{\mathcal{Y}} f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(y))), s) dG_{y|s}(y|s) dG_s(s) \quad (22)$$

where $\mathcal{Y} = [\underline{y}, \bar{y}] = [\underline{s} + \underline{e}, \bar{s} + \bar{e}]$. The conditional distribution $G_{y|s}$ can be written, using law of total probability, as²⁵

$$G_{y|s}(x|s) = \Pr\{y \leq x|s\} = p \Pr\{s + e \leq x|s\} + (1 - p) \Pr\{h(s + e) \leq x|s\} \quad (23)$$

$$= pG_{y_1|s}(x|s) + (1 - p)G_{y_1|s}(h^{-1}(x)|s) \quad (24)$$

So we have the decomposition $V_L = V_{A,1} + V_{A,2}$ where

$$V_{A,1} = \int_{\mathcal{S}} \int_{\mathcal{Y}} p \cdot f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(y))), s) dG_{y_1|s}(y|s) dG_s(s) \quad (25)$$

$$V_{A,2} = \int_{\mathcal{S}} \int_{\mathcal{Y}} (1 - p) \cdot f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(y))), s) dG_{y_1|s}(h^{-1}(y)|s) dG_s(s) \quad (26)$$

With a change of variables $y_1 := h^{-1}(y)$ the second integral becomes

$$V_{A,2} = \int_{\mathcal{S}} \int_{\mathcal{Y}} (1 - p) \cdot f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(y))), s) g_{y_1|s}(h^{-1}(y)|s) (h^{-1})'(y) dy dG_s(s) \quad (27)$$

$$= \int_{\mathcal{S}} \int_{\mathcal{Y}} (1 - p) \cdot f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(h(y)))), s) g_{y_1|s}(z|s) (h'(z))^{-1} h'(z) dz dG_s(s) \quad (28)$$

$$= \int_{\mathcal{S}} \int_{\mathcal{Y}} (1 - p) \cdot f(G_x^{-1}(G_{\bar{s}}(\tilde{s}(h(y_1)))), s) dG_{y_1|s}(y_1|s) dG_s(s) \quad (29)$$

By concavity of f ,

$$V_L \leq \int_{\mathcal{S}} \int_{\mathcal{Y}} f(\sigma_A(y), s) dG_{y_1|s}(y|s) dG_s(s) =: I \quad (30)$$

²⁵Note that we have used same decomposition to derive $\mathbb{E}[s|Y_1 = h^{-1}(y)] = \hat{s}(h^{-1}(y))$ in the main text. To see this, consider $\mathbb{E}[s|Y_2 = y] := \int s g_{s|y_2}(s|y) ds$, and the CDF of y_2 can be written as $G_{y_2}(x) = \Pr\{Y_2 \leq x\} = \Pr\{h(s + e) \leq x\} = \Pr\{y_1 \leq h^{-1}(x)\} = G_{y_1}(h^{-1}(x))$. Similarly $G_{y_2|s}(x|s) = G_{y_1}(h^{-1}(x)|s)$. Then by Bayes' rule:

$$g_{s|y_2}(s|y) = \frac{g_{y_2|s}(y|s) g_s(s)}{g_y} = \frac{g_{y_1|s}(h^{-1}(y)|s) g_s(s)}{g_{y_1}(h^{-1}(y))} \equiv g_{s|y_1}(s|h^{-1}(y))$$

Where

$$\sigma_A(y) = pG_x^{-1}G_{\hat{s}}(\tilde{s}(y)) + (1 - p)G_x^{-1}G_{\hat{s}}(\tilde{s}(h(y)))$$

Note that V_I also can be written as

$$V_I = \int_{\mathcal{S}} \int_{\mathcal{Y}} f(\sigma_I(y), s) dG_{y_1|s}(y|s) dG_s(s) \quad (31)$$

where $\sigma_I(y) = G_x^{-1}(G_{\hat{s}}(\hat{s}(y)))$. Note that since the integral I in (30) is with respect to the joint distribution of y_1 and s , it provides an upper bound on the value of the economy with AI in terms of an economy without AI (but with imperfect information in terms of y_1) with a different assignment function σ_A . However since σ_I is the unique monotonically-increasing 1-to-1 assignment in such an economy and $\sigma_A \neq \sigma_I$ a.s., we have by Proposition 2 that V_I must be larger than V_L . ■

E.4 LLM Adoption Correlated With Ability

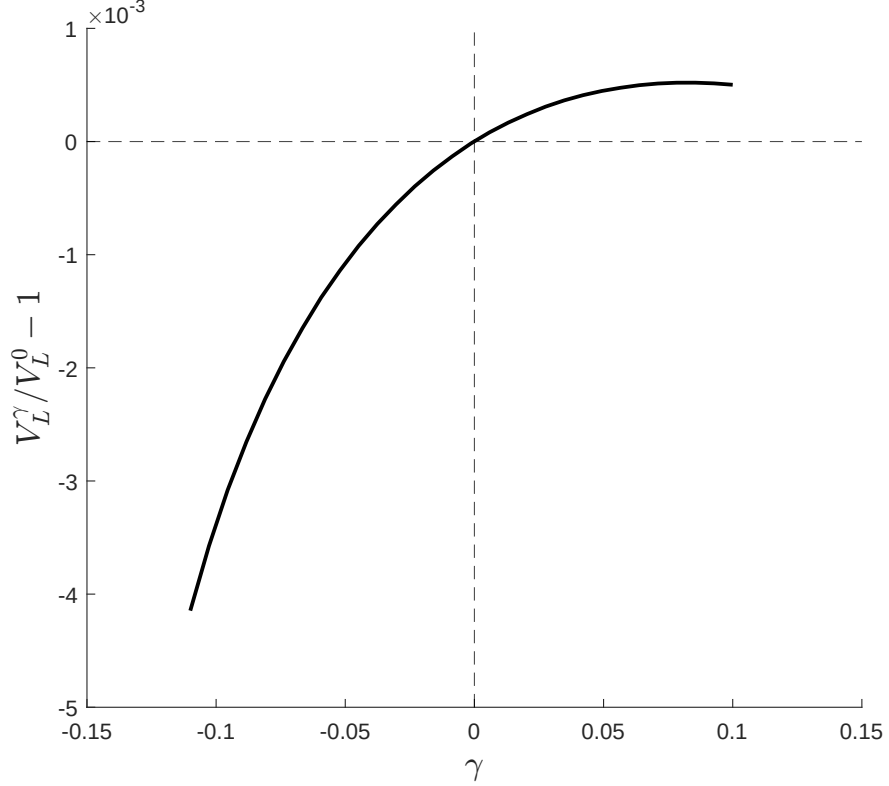
In this extension with model p as a linear function of underlying ability s , thus making the coin-flip of whether or not an applicant uses LLMs correlated with s . We write this function as:

$$p = \psi(s) := p_0 - \gamma s \quad (32)$$

Such that higher γ means *higher* correlation of underlying ability with usage of LLM (lower p). This equation implies that the average probability of adoption in the cross section is therefore equal to $p_0 - \gamma\mu_s$. In our counterfactual exercise of varying γ we fix p_0 such that this average adoption remains constant at 0.5. Therefore we set $p_0 = 0.5 + \gamma\mu_s$.

Figure A16 shows total output when varying γ between negative and positive values as compared to the baseline model where all workers use LLM with the same probability, i.e. $\gamma = 0$ and $p = p_0 = 0.5$.

Figure A16. Sensitivity of output to correlation between LLM use and skill



Notes: This figure calculates the changes in output in the economy with imperfect information and LLM technology when varying the relationship between probability of LLM adoption and worker skill, i.e. varying γ with higher γ denoting higher correlation of LLM adoption with skill. The output where the use of LLM and skill are uncorrelated ($\gamma = 0$ and thus $p = 0.5$ for everyone) is chosen as the comparison benchmark.